15 Trigonometry II (10A)

LESSON SEQUENCE

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15.2 The sine rule 1035
15.3 The cosine rule 1045
15.4 Area of triangles 1051
15.5 The unit circle 1057
15.6 Trigonometric functions 1064
15.7 Solving trigonometric equations 1072
15.8 Review 1078



LESSON
15.1 Overview

Why learn this?



Trigonometry is the branch of mathematics that describes the relationship between angles and side lengths in triangles. The ability to calculate distances using angles has long been critical. As early as the third century BCE, trigonometry was being used in the study of astronomy. Early explorers, using rudimentary calculations and the stars, were able to navigate their way around the world. They were even able to map coastlines along the way. Cartographers use trigonometry when they are making maps. It is essential to be able to calculate distances that can't be physically measured. Astronomers use trigonometry to calculate distances such as that from a particular planet to Earth. Our explorations have now turned towards the skies and outer space.

Scientists design and launch space shuttles and rockets to explore our universe. By applying trigonometry, they can approximate the distances to other planets. As well as in astronomy and space exploration, trigonometry is widely used in many other areas. Surveyors use trigonometry in setting out a land subdivision. Builders, architects and engineers use angles, lengths and forces in the design and construction of all types of buildings, both domestic and industrial. In music, a single note is a sine wave. Sound engineers manipulate sine waves to create sound effects. Trigonometry has many real-life applications.

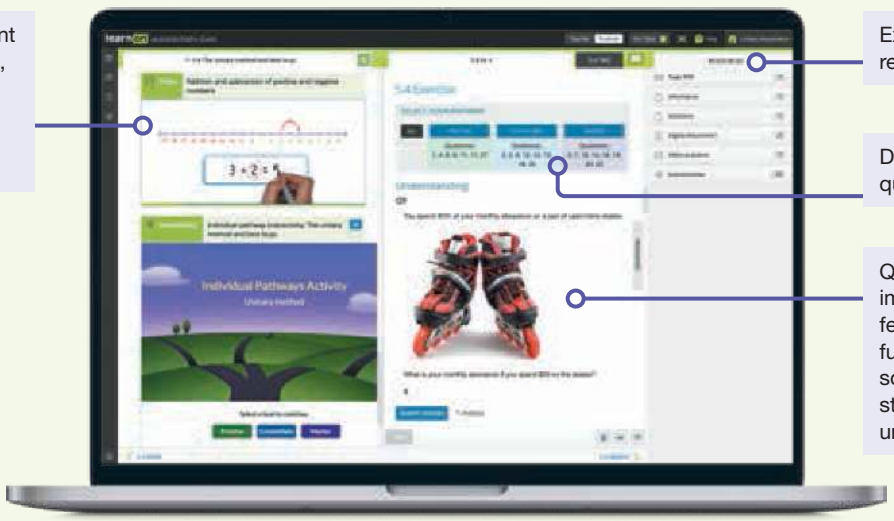
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Extra learning resources

Differentiated question sets

Questions with immediate feedback, and fully worked solutions to help students get unstuck

Exercise 15.1 Pre-test

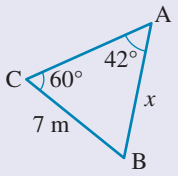
learnon

1. MC From the following options select the exact value of sin(30°).

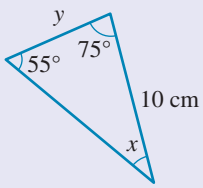
A. $\frac{1}{\sqrt{2}}$ **B.** $\frac{1}{2}$

C. $\frac{\sqrt{3}}{2}$ **D.** 1

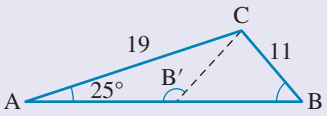
2. Solve for x, correct to 2 decimal places.



3. Solve for y, correct to 2 decimal places.



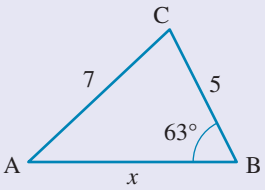
4. MC Choose the values of the angles B and B′ in the below triangle, to the nearest degree. (Assume $BC = B'C$.)



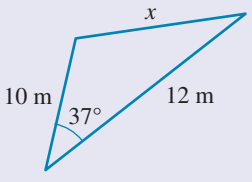
A. 47° and 133° **B.** 46° and 134°

C. 47° and 153° **D.** 25° and 155°

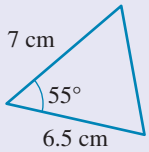
5. Calculate the perimeter of the following triangle, correct to 2 decimal places.



6. Solve for x, correct to 1 decimal place.



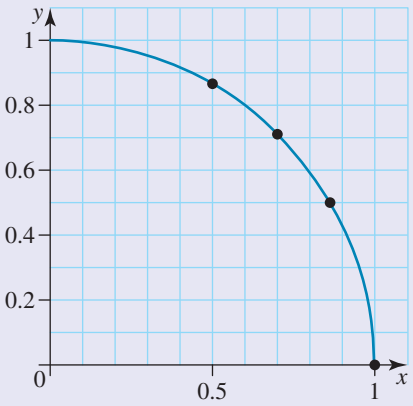
7. MC Calculate the area of the triangle shown.



A. 13.05 cm^2 **B.** 18.64 cm^2
C. 22.75 cm^2 **D.** 26.1 cm^2

8. State in which quadrant of the unit circle is the angle 203° located.

9. Determine the value of cos(60°) using part of the unit circle.



10. MC If $\cos(x°) = p$ for $0 \leq x \leq 90°$, then $\sin(180 - x°)$ in terms of p is:

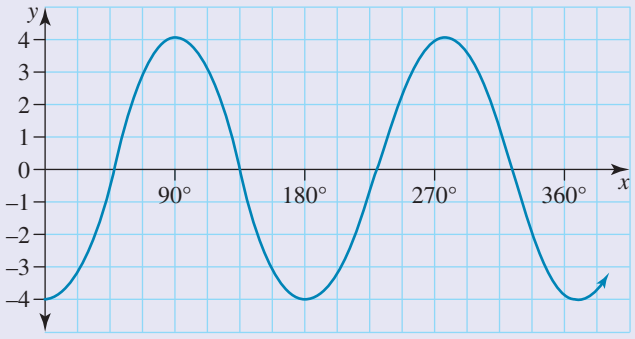
A. p **B.** $180 - p$ **C.** $1 - p$ **D.** $\sqrt{1 - p^2}$

11. MC Select the amplitude and period, respectively, of $y = -2\sin(2x)$ from the following options.

A. $-2, 360°$ **B.** $-2, 180°$ **C.** $2, 2$ **D.** $2, 180°$

12. MC Select the correct equation for the graph shown.

A. $y = 4\cos(2x)$
B. $y = -4\cos(2x)$
C. $y = -4\sin(2x)$
D. $y = 4\sin(2x)$



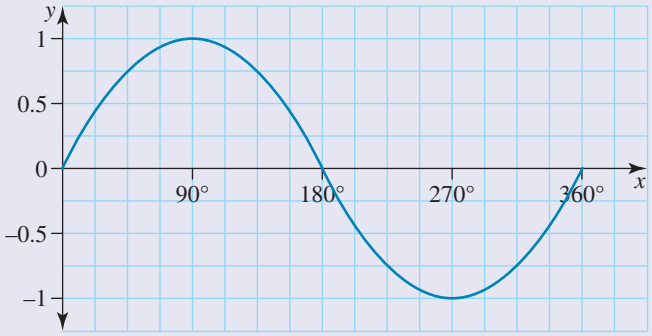
13. MC Select the correct solutions for the equation $\sin(x) = \frac{1}{2}$ for x over the domain $0 \leq x \leq 360°$.

A. $x = 30°$ and $x = 150°$ **B.** $x = 30°$ and $x = 210°$
C. $x = 60°$ and $x = 120°$ **D.** $x = 60°$ and $x = 240°$

14. MC Select the correct solutions for the equation $\cos(2x) = -\frac{\sqrt{2}}{2}$ for x over the domain $0 \leq x \leq 360°$.

A. $x = 22.5°$ and $x = 337.5°$ **B.** $x = 45°$ and $x = 315°$
C. $x = 67.5°$ and $x = 112.5°$ **D.** $x = 157.5°$ and $x = 202.5°$

15. Using the graph shown, solve the equation $7\sin(x) = -7$ for $0 \leq x \leq 360°$.



LESSON
15.2 The sine rule

LEARNING INTENTIONS

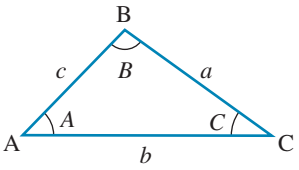
At the end of this lesson you should be able to:

- apply the sine rule to evaluate angles and sides of triangles
- recognise when the ambiguous case of the sine rule exists.

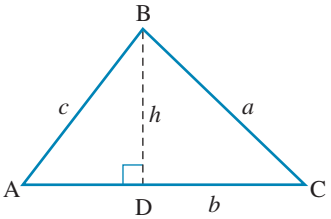
15.2.1 The sine rule

eles-3214

- In any triangle, label the angles are named by the vertices A, B and C and the corresponding opposite sides as a, b and c as shown in the diagram.



- Let BD be the perpendicular line from B to AC, of length h, giving two right-angled triangles, ΔADB and ΔCDB.



Using ΔADB:

$$\sin(A) = \frac{h}{c}$$
$$h = c\sin(A)$$

Using ΔCDB:

$$\sin(C) = \frac{h}{a}$$
$$h = a\sin(C)$$

Equating the values of h:

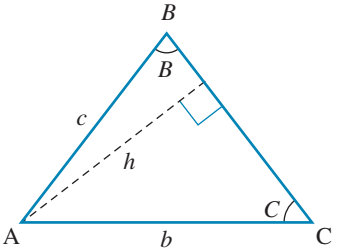
$$c\sin(A) = a\sin(C)$$

giving:

$$\frac{c}{\sin(C)} = \frac{a}{\sin(A)}$$

$\frac{h}{c} = \sin(A)$ and $\frac{h}{a} = \sin(C)$

- Similarly, if a perpendicular line is drawn from vertex A to BC, then:



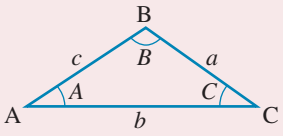
$$\frac{c}{\sin(C)} = \frac{b}{\sin(B)}$$

$h = c\sin(B)$ and $h = b\sin(C)$

Sine rule

- **The sine rule for any triangle ABC is:**

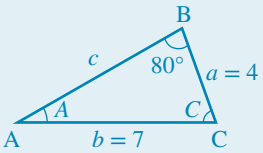
$$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$$



- The sine rule can be used to solve non-right-angled triangles if we are given:
  1. two angles and one side
  2. two sides and an angle opposite one of these sides.

WORKED EXAMPLE 1 Determining unknown angles and sides of a given triangle

In the triangle ABC, $a = 4$ m, $b = 7$ m and $B = 80°$. Calculate the values of A, C and c.

THINK	WRITE/DRAW
1. Draw a labelled diagram of the triangle ABC and fill in the given information.	
2. Check that one of the criteria for the sine rule has been satisfied.	The sine rule can be used since two side lengths and an angle opposite one of these side lengths have been given.
3. Write down the sine rule to calculate A.	To calculate angle A: $\frac{a}{\sin(A)} = \frac{b}{\sin(B)}$
4. Substitute the known values into the rule.	$\frac{4}{\sin(A)} = \frac{7}{\sin(80°)}$
5. Transpose the equation to make $\sin(A)$ the subject.	$4\sin(80°) = 7\sin(A)$ $\sin(A) = \frac{4\sin(80°)}{7}$
6. Evaluate and write your answer.	$A = \sin^{-1}\left(\frac{4\sin(80°)}{7}\right)$ $\approx 34.246\,004\,71°$
7. Round off the answer to degrees and minutes.	$\approx 34°15'$
8. Determine the value of angle C using the fact that the angle sum of any triangle is 180°.	$C \approx 180° - (80° + 34°15')$ $= 65°45'$
9. Write down the sine rule to calculate the value of c.	To calculate side length c: $\frac{c}{\sin(C)} = \frac{b}{\sin(B)}$
10. Substitute the known values into the rule.	$\frac{c}{\sin(65°45')} = \frac{7}{\sin(80°)}$

11. Transpose the equation to make c the subject. $c = \dfrac{7\sin(65°45')}{\sin(80°)}$

12. Evaluate. Round off the answer to 2 decimal places and include the appropriate unit. ≈ 6.48 m

TI | THINK

Open a new document and a Calculator page. Ensure your calculator is set to the degree and approximate mode, as shown for the next set of examples.
To do this, press:
- HOME
- 5: Settings
- 2: Document Settings

For Display Digits select 'Fix 2'.
TAB to Angle and select 'Degree'.
TAB to Calculation Mode and select 'Approximate'.
TAB to OK and press ENTER.

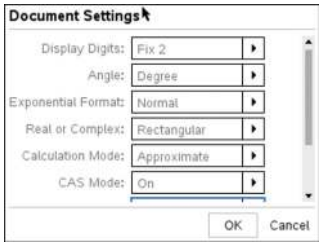
1. In a new problem, on the Calculator page, press TRIG to access and select the appropriate trigonometric ratio $(\sin^{-1})$. Complete the entry line as:

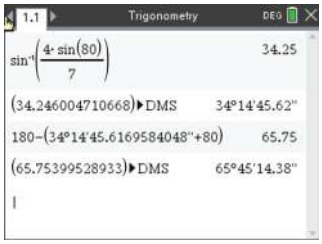
$$\sin^{-1}\left(\frac{4\sin(80)}{7}\right)$$

Then press ENTER.
To convert the decimal degree into degrees, minutes and seconds, press:
- CATALOG
- 1
- D

Scroll to and select ▶ DMS then press ENTER.
Complete the entry line as: 180 − (ans + 80)
Then press ENTER.
Repeat the above process to convert to degrees, minutes and seconds.

DISPLAY/WRITE





$A = 34°15'$ and $C = 65°45'$

CASIO | THINK

Ensure your calculator is set to the degree and decimal modes. To do this, at the bottom of the Main screen, tap on the mode options until you have Decimal and Deg.

1. On the Main screen, complete the entry line as:

$$\sin^{-1}\left(\frac{4\sin(80)}{7}\right)$$

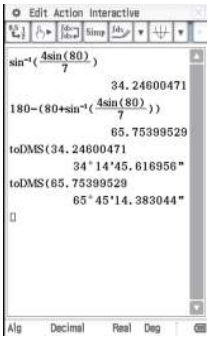
$$180 - \left(80 + \sin^{-1}\left(\frac{4\sin(80)}{7}\right)\right)$$

Press EXE after each entry.
To convert the decimal degree into degrees, minutes and seconds, tap:
- Action
- Transformation
- DMS
- toDMS

Highlight and drag each of the decimal answers into the entry line and press EXE.

DISPLAY/WRITE



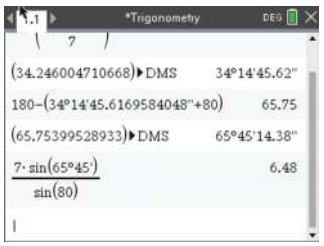


$A = 34°15'$ and $C = 65°45'$ rounded to the nearest minute.

2. To find the value of c, complete the entry line as:

$$\frac{7\sin(\text{ans})}{\sin 80}$$

Then press ENTER.

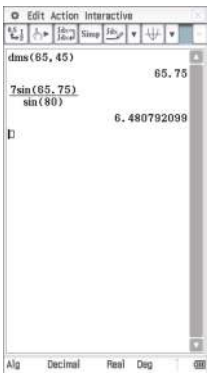


$c = 6.48$ m

2. To calculate the value of c, complete the entry lines as:
dms(65, 45)

$$\frac{7\sin(65, 75)}{\sin(80)}$$

Press EXE after each entry line.

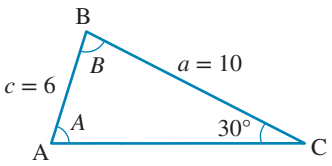


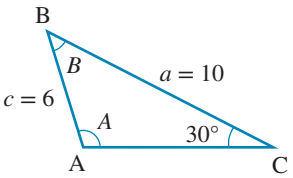
$c = 6.48$ m

eles-5005

15.2.2 The ambiguous case

- If two side lengths and an angle opposite one of these side lengths are given, then two different triangles may be drawn.
- For example, if $a = 10$, $c = 6$ and $C = 30°$, two possible triangles could be created.
- In the first case angle A is an acute angle, and in the second case angle A is an obtuse angle.
- When using the sine rule to determine an angle, the inverse sine function is used.
- In lesson 15.5, you will see that the sine of an angle between 0° and 90° has the same value as the sine of its supplement.
  For example, $\sin 40° \approx 0.6427$ and $\sin 140° \approx 0.6427$.





WORKED EXAMPLE 2 Solving triangles and checking for the ambiguous case

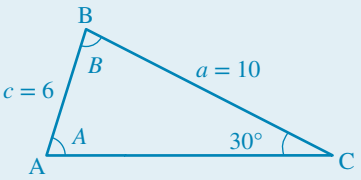
In the triangle ABC, $a = 10$ m, $c = 6$ m and $C = 30°$. Determine two possible values of A, and hence two possible values of B and b.

Case 1

THINK

1. Draw a labelled diagram of the triangle ABC and fill in the given information.

WRITE/DRAW



2. Check that one of the criteria for the sine rule has been satisfied.

The sine rule can be used since two side lengths and an angle opposite one of these side lengths have been given.

3. Write down the sine rule to determine A.

To determine angle A: $\dfrac{a}{\sin(A)} = \dfrac{c}{\sin(C)}$

4. Substitute the known values into the rule.

$$\frac{10}{\sin(A)} = \frac{6}{\sin(30°)}$$

$$10\sin(30°) = 6\sin(A)$$

5. Transpose the equation to make $\sin(A)$ the subject.

$$\sin(A) = \frac{10\sin(30°)}{6}$$

6. Evaluate angle A.

$$A = \sin^{-1}\left(\frac{10\sin(30°)}{6}\right)$$
$$\approx 56.442\,690\,24°$$

7. Round off the answer to degrees and minutes.

$$A = 56°27'$$

8. Determine the value of angle B, using the fact that the angle sum of any triangle is 180°.

$$B \approx 180° - (30° + 56°27')$$
$$= 93°33'$$

9. Write down the sine rule to calculate b.

To calculate side length b:

$$\frac{b}{\sin(B)} = \frac{c}{\sin(C)}$$

10. Substitute the known values into the rule.

$$\frac{b}{\sin(93°33')} = \frac{6}{\sin(30°)}$$

11. Transpose the equation to make b the subject.

$$b = \frac{6\sin(93°33')}{\sin(30°)}$$

12. Evaluate. Round off the answer to 2 decimal places and include the appropriate unit.

$$\approx 11.98\text{ m}$$

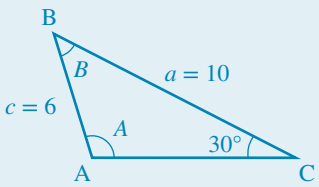
Note: The values we have just obtained are only one set of possible answers for the given dimensions of the triangle ABC.
We are told that $a = 10$ m, $c = 6$ m and $C = 30°$. Since side a is larger than side c, it follows that angle A will be larger than angle C. Angle A must be larger than 30°; therefore it may be an acute angle or an obtuse angle.

Case 2

THINK

1. Draw a labelled diagram of the triangle ABC and fill in the given information.

WRITE/DRAW



2. Write down the alternative value for angle A. Simply subtract the value obtained for A in case 1 from 180°.

To determine the alternative angle A:
If $\sin A = 0.8333$, then A could also be:
$A \approx 180° - 56°27'$
$= 123°33'$

3. Determine the alternative value of angle B, using the fact that the angle sum of any triangle is 180°.

$B \approx 180° - (30° + 123°33')$
$= 26°27'$

4. Write down the sine rule to determine the alternative b.

To calculate side length b:

$$\frac{b}{\sin(B)} = \frac{c}{\sin(C)}$$

5. Substitute the known values into the rule. $\frac{b}{\sin(26°27')} = \frac{6}{\sin(30°)}$

6. Transpose the equation to make b the subject. $b = \frac{6\sin(26°27')}{\sin(30°)}$

7. Evaluate. Round off the answer to 2 decimal places and include the appropriate unit. ≈ 5.34 m

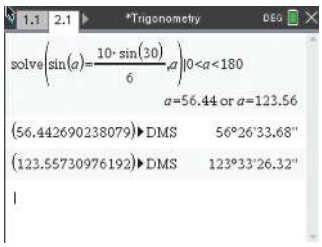
TI | THINK

1. In a new problem, on a Calculator page, complete the entry line as:
   solve
   $\left(\sin(a) = \frac{10\sin(30)}{6}, a\right) |$
   $0 < a < 180$
   Press the up arrow and highlight the answer shown. Press ENTER to bring this answer down into the new line.
   Convert these angles to degrees and minutes as shown in Worked Example 1 by completing the entry as shown.
   Press ENTER after each entry.

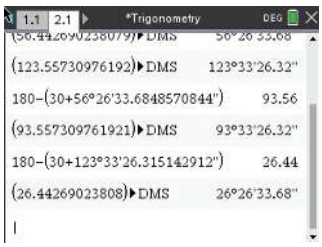
2. Solve for the two values of B as shown in the screenshot. Instead of typing the angle manually, press the up arrow to highlight the previous answer you want, then press ENTER. Press ENTER after each entry.
   Convert these angles to degrees and minutes.

3. To solve for the two values of b, complete the entry lines as:
   $\frac{6\sin(93°34')}{\sin(30)}$
   $\frac{6\sin(26°26')}{\sin(30)}$
   Press ENTER after each entry.

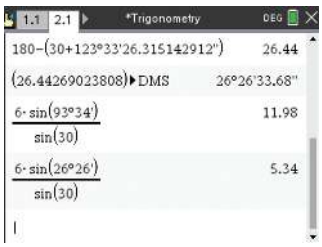
DISPLAY/WRITE



$A = 56°27'$ or $123°34'$



$B = 93°33'$ or $26°27'$



$b = 11.98$ m or 5.34 m

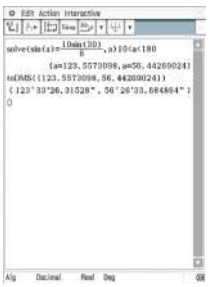
CASIO | THINK

1. On the Main screen, complete the entry line as:
   solve
   $\left(\sin(a) = \frac{10\sin(30)}{6}, a\right) |$
   $0 < a < 180$
   Convert these angles to degrees and minutes as shown.
   Press EXE after each entry.

2. Solve for the two values of B as shown at right. [This uses $B = 180 - (30 + A)$.]
   Press EXE after each entry, and convert these angles to degrees and minutes.

3. To solve for the two values of b, complete the entry lines as:
   $\frac{6\sin(93.55730976)}{\sin(30)}$
   $\frac{6\sin(26.4426902)}{\sin(30)}$
   Press after each entry.

DISPLAY/WRITE



$A = 56°27'$ or $123°33'$ rounded to the nearest minute.

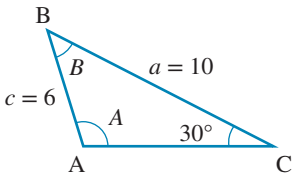


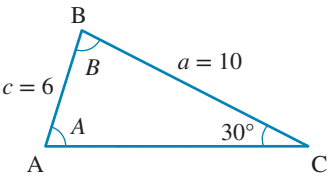
$B = 93°33'$ or $26°27'$



$b = 11.98$ m or 5.34 m

- In Worked example 2 there were two possible solutions, as shown by the diagrams below.





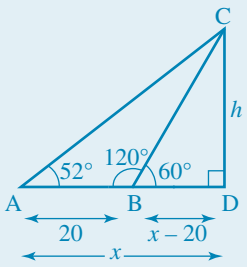
- The ambiguous case does not apply to every question.

  Consider Worked example 1.
  - Since $\angle A = 34°15'$, it could also have been $\angle A = 145°45'$, the supplementary angle.
  - If $\angle A = 34°15'$ and $\angle B = 80°$, then
    $\angle C = 65°45'$ (angle sum of triangle).
  - If $\angle A = 145°45'$ and $\angle B = 80°$, then
    $\angle C = 180° - (145°45' + 80°)$ which is not possible.
    $\angle C = -45°45'$

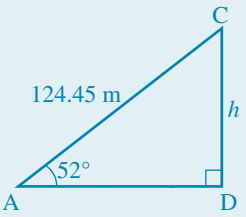
  Hence, for Worked example 1, only one possible solution exists.
- The ambiguous case may exist if the angle found is opposite the larger given side.

WORKED EXAMPLE 3 Calculating heights given angles of elevation

To calculate the height of a building, Kevin measures the angle of elevation to the top as 52°. He then walks 20 m closer to the building and measures the angle of elevation as 60°. Calculate the height of the building.

THINK	WRITE/DRAW
1. Draw a labelled diagram of the situation and fill in the given information.	
2. Check that one of the criteria for the sine rule has been satisfied for triangle ABC.	The sine rule can be used for triangle ABC since two angles and one side length have been given.
3. Calculate the value of angle ACB, using the fact that the angle sum of any triangle is 180°.	$\angle ACB = 180° - (52° + 120°)$ $= 8°$
4. Write down the sine rule to calculate b (or AC).	To calculate side length b of triangle ABC: $\frac{b}{\sin(B)} = \frac{c}{\sin(C)}$
5. Substitute the known values into the rule.	$\frac{b}{\sin(120°)} = \frac{20}{\sin(8°)}$
6. Transpose the equation to make b the subject.	$b = \frac{20\sin(120°)}{\sin(8°)}$
7. Evaluate. Round off the answer to 2 decimal places and include the appropriate unit.	≈ 124.45 m

8. Draw a diagram of the situation, that is triangle ADC, labelling the required information.
Note: There is no need to solve the rest of the triangle in this case as the values will not assist in calculating the height of the building.



9. Write down what is given for the triangle. — Have: angle and hypotenuse
10. Write down what is needed for the triangle. — Need: opposite side
11. Determine which of the trigonometric ratios is required (SOH – CAH – TOA). — $\sin(\theta) = \frac{O}{H}$
12. Substitute the given values into the appropriate ratio. — $\sin(52°) = \frac{h}{124.45}$
13. Transpose the equation and solve for h. — $124.45 \sin(52°) = h$, $h = 124.45 \sin(52°)$
14. Round off the answer to 2 decimal places. — $h \approx 98.07$
15. Write the answer. — The height of the building is 98.07 m.

DISCUSSION

Discuss a real-life situation in which the sine rule can be used.

Resources

eWorkbook Topic 15 Workbook (worksheets, code puzzle and project) (ewbk-13308)
Interactivities The ambiguous case (int-4818)
The sine rule (int-6275)

Exercise 15.2 The sine rule

15.2 Quick quiz | **15.2 Exercise**

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 2, 6, 7, 11, 13, 17, 18, 21	3, 4, 8, 9, 12, 14, 16, 19, 22	5, 10, 15, 20, 23, 24, 25

Where appropriate in this exercise, write your angles correct to the nearest minute and side lengths correct to 2 decimal places.

Fluency

1. WE1 In the triangle ABC, $a = 10$, $b = 12$ and $B = 58°$. Calculate A, C and c.
2. In the triangle ABC, $c = 17.35$, $a = 26.82$ and $A = 101°47'$. Calculate C, B and b.

3. In the triangle ABC, $a = 5$, $A = 30°$ and $B = 80°$. Calculate C, b and c.

4. In the triangle ABC, $c = 27$, $C = 42°$ and $A = 105°$. Calculate B, a and b.

5. In the triangle ABC, $a = 7$, $c = 5$ and $A = 68°$. Determine the perimeter of the triangle.

6. Calculate all unknown sides and angles for the triangle ABC, given $A = 57°$, $B = 72°$ and $a = 48.2$.

7. Calculate all unknown sides and angles for the triangle ABC, given $a = 105$, $B = 105°$ and $C = 15°$.

8. Calculate all unknown sides and angles for the triangle ABC, given $a = 32$, $b = 51$ and $A = 28°$.

9. Calculate the perimeter of the triangle ABC if $a = 7.8$, $b = 6.2$ and $A = 50°$.

10. MC In a triangle ABC, $B = 40°$, $b = 2.6$ and $c = 3$. Identify the approximate value of C.
Note: There may be more than one correct answer.

A. 47° **B.** 48° **C.** 132° **D.** 133°

Understanding

11. WE2 In the triangle ABC, $a = 10$, $c = 8$ and $C = 50°$. Determine two possible values of A and hence two possible values of b.

12. In the triangle ABC, $a = 20$, $b = 12$ and $B = 35°$. Determine two possible values for the perimeter of the triangle.

13. Calculate all unknown sides and angles for the triangle ABC, given $A = 27°$, $B = 43°$ and $c = 6.4$.

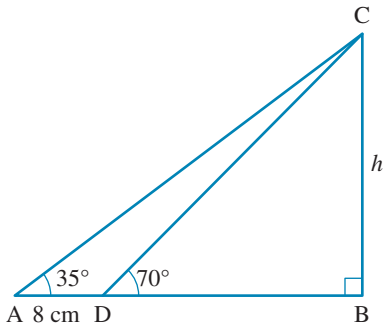
14. Calculate all unknown sides and angles for the triangle ABC, given $A = 100°$, $b = 2.1$ and $C = 42°$.

15. Calculate all unknown sides and angles for the triangle ABC, given $A = 25°$, $b = 17$ and $a = 13$.

16. WE3 To calculate the height of a building, Kevin measures the angle of elevation to the top as 48°. He then walks 18 m closer to the building and measures the angle of elevation as 64°. Calculate the height of the building.

Reasoning

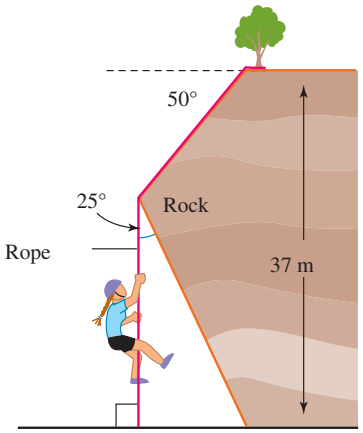
17. Calculate the value of h, correct to 1 decimal place. Show the full working.



18. A boat sails on a bearing of N15°E for 10 km and then on a bearing of S85°E until it is due east of the starting point. Determine the distance from the starting point to the nearest kilometre. Show all your working.

19. A hill slopes at an angle of 30° to the horizontal. A tree that is 8 m tall and leaning downhill is growing at an angle of 10° m to the vertical and is part-way up the slope. Evaluate the vertical height of the top of the tree above the slope. Show all your working.

20. A cliff is 37 m high. The rock slopes outward at an angle of 50° to the horizontal and then cuts back at an angle of 25° to the vertical, meeting the ground directly below the top of the cliff. Carol wishes to abseil from the top of the cliff to the ground as shown in the diagram.



Her climbing rope is 45 m long and is secured to a tree at the top of the cliff. Determine if the rope will be long enough to allow her to reach the ground.

Problem solving

21. A river has parallel banks that run directly east–west. From the south bank, Kylie takes a bearing to a tree on the north side. The bearing is 047°T. She then walks 10 m due east and takes a second bearing to the tree. This is 305°T.
Determine:

a. her distance from the second measuring point to the tree
b. the width of the river, to the nearest metre.

22. A ship sails on a bearing of S20°W for 14 km; then it changes direction, sails for 20 km and drops anchor. Its bearing from the starting point is now N65°W.

a. Determine the distance of the ship from its starting point.
b. Calculate the bearing on which the ship sails for the 20 km leg.

23. A cross-country runner runs at 8 km/h on a bearing of 150°T for 45 minutes; then she changes direction to a bearing of 053°T and runs for 80 minutes at a different speed until she is due east of the starting point.

a. Calculate the distance of the second part of the run.
b. Calculate her speed for this section, correct to 1 decimal place.
c. Evaluate how far she needs to run to get back to the starting point.



24. From a fire tower, A, a fire is spotted on a bearing of N42°E. From a second tower, B, the fire is on a bearing of N12°W. The two fire towers are 23 km apart, and A is N63°W of B.
Determine how far the fire is from each tower.

25. A yacht sets sail from a marina and sails on a bearing of 065°T for 3.5 km. It then turns and sails on a bearing of 127°T for another 5 km.

a. Evaluate the distance of the yacht from the marina, correct to 1 decimal place.
b. If the yacht was to sail directly back to the marina, on what bearing should it travel?
Give your answer rounded to the nearest minute.



LESSON
15.3 The cosine rule

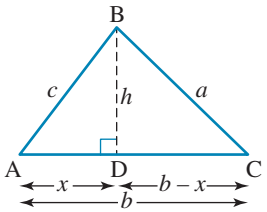
LEARNING INTENTIONS

At the end of this lesson you should be able to:
- apply the cosine rule to calculate a side of a triangle
- apply the cosine rule to calculate the angles of a triangle.

15.3.1 The cosine rule

eles-5006

- In triangle ABC, let BD be the perpendicular line from B to AC, of length h, giving two right-angled triangles, ΔADB and ΔCDB.
- Let the length of AD $= x$, then DC $= (b - x)$.
- Using triangle ADB and Pythagoras' theorem, we obtain:



$$c^2 = h^2 + x^2 \qquad [1]$$

Using triangle CDB and Pythagoras' theorem, we obtain:

$$a^2 = h^2 + (b - x)^2 \qquad [2]$$

Expanding the brackets in equation [2]:

$$a^2 = h^2 + b^2 - 2bx + x^2$$

Rearranging equation [2] and using $c^2 = h^2 + x^2$ from equation [1]:

$$\begin{aligned} a^2 &= h^2 + x^2 + b^2 - 2bx \\ &= c^2 + b^2 - 2bx \\ &= b^2 + c^2 - 2bx \end{aligned}$$

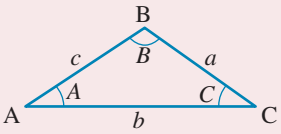
From triangle ABD, $x = c\cos(A)$; therefore, $a^2 = b^2 + c^2 - 2bx$ becomes

$$a^2 = b^2 + c^2 - 2bc\cos(A).$$

Cosine rule

- **The cosine rule for any triangle ABC is:**

$$\boldsymbol{a^2 = b^2 + c^2 - 2bc\cos(A)}$$
$$\boldsymbol{b^2 = a^2 + c^2 - 2ac\cos(B)}$$
$$\boldsymbol{c^2 = a^2 + b^2 - 2ab\cos(C)}$$



- The cosine rule can be used to solve non-right-angled triangles if we are given:
  1. three sides
  2. two sides and the included angle.

Note: Once the third side has been calculated, the sine rule can be used to determine other angles if necessary.

Calculating angles by transposing the cosine rule

- **If three sides of a triangle are known, an angle can be found by transposing the cosine rule to make $\cos(A)$, $\cos(B)$ or $\cos(C)$ the subject.**

$$a^2 = b^2 + c^2 - 2bc\cos(A) \Rightarrow \cos(A) = \frac{b^2 + c^2 - a^2}{2bc}$$

$$b^2 = a^2 + c^2 - 2ac\cos(B) \Rightarrow \cos(B) = \frac{a^2 + c^2 - b^2}{2ac}$$

$$c^2 = a^2 + b^2 - 2ab\cos(C) \Rightarrow \cos(C) = \frac{a^2 + b^2 - c^2}{2ab}$$

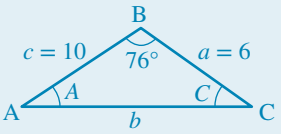
WORKED EXAMPLE 4 Calculating sides using the cosine rule

Calculate the third side of triangle ABC given $a = 6$, $c = 10$ and $B = 76°$.

THINK

1. Draw a labelled diagram of the triangle ABC and fill in the given information.
2. Check that one of the criteria for the cosine rule has been satisfied.
3. Write down the appropriate cosine rule to calculate side b.
4. Substitute the given values into the rule.
5. Evaluate.
6. Round off the answer to 2 decimal places.

WRITE/DRAW



Yes, the cosine rule can be used since two side lengths and the included angle have been given.

To calculate side b:

$b^2 = a^2 + c^2 - 2ac\cos(B)$

$= 6^2 + 10^2 - 2 \times 6 \times 10 \times \cos(76°)$

$\approx 106.969\,372\,5$

$b \approx \sqrt{106.969\,372\,5}$

≈ 10.34

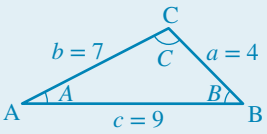
WORKED EXAMPLE 5 Calculating angles using the cosine rule

Calculate the smallest angle in the triangle with sides 4 cm, 7 cm and 9 cm.

THINK

1. Draw a labelled diagram of the triangle, call it ABC and fill in the given information.
   Note: The smallest angle will correspond to the smallest side.
2. Check that one of the criteria for the cosine rule has been satisfied.

WRITE/DRAW



Let $a = 4, b = 7, c = 9$

The cosine rule can be used since three side lengths have been given.

3. Write down the appropriate cosine rule to calculate angle A.

$$\cos(A) = \frac{b^2 + c^2 - a^2}{2bc}$$

4. Substitute the given values into the rearranged rule.

$$= \frac{7^2 + 9^2 - 4^2}{2 \times 7 \times 9}$$

5. Evaluate.

$$= \frac{114}{126}$$

6. Transpose the equation to make A the subject by taking the inverse cos of both sides.

$$A = \cos^{-1}\left(\frac{114}{126}\right)$$
$$\approx 25.208\,765\,3°$$

7. Round off the answer to degrees and minutes.

$$\approx 25°13'$$

TI | THINK

In a new problem, on a Calculator page, complete the entry lines as:

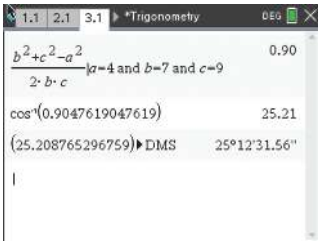
$\frac{b^2 + c^2 - a^2}{2bc} | a = 4$ and $b = 7$ and $c = 9$

$\cos^{-1}$ (ans)

Convert the angle to DMS as shown previously.

Press ENTER after each entry.

DISPLAY/WRITE



The smallest angle is $25°13'$ rounded up to the nearest minute.

CASIO | THINK

On the Main screen, complete the entry lines as:

$\frac{b^2 + c^2 - a^2}{2b \times c} | a = 4 | b = 7 | c = 9$

$\cos^{-1}$ (0.9047619048)

Convert the angle to DMS as shown previously.

Press EXE after each entry.

DISPLAY/WRITE



The smallest angle, rounded up to the nearest minute, is $25°13'$.

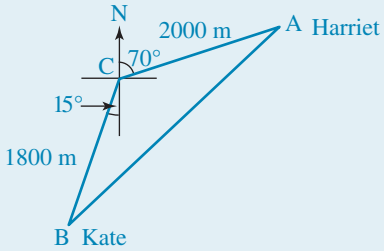
WORKED EXAMPLE 6 Applying the cosine rule to solve problems

Two rowers, Harriet and Kate, set out from the same point. Harriet rows N70°E for 2000 m and Kate rows S15°W for 1800 m.
Calculate the distance between the two rowers, correct to 2 decimal places.

THINK

1. Draw a labelled diagram of the triangle, call it ABC and fill in the given information.

WRITE/DRAW



2. Check that one of the criteria for the cosine rule has been satisfied.

The cosine rule can be used since two side lengths and the included angle have been given.

3. Write down the appropriate cosine rule to calculate side c.

To calculate side c:

$c^2 = a^2 + b^2 - 2ab\cos(C)$

4. Substitute the given values into the rule. $= 2000^2 + 1800^2 - 2 \times 2000 \times 1800 \cos(125°)$

5. Evaluate. $\approx 11\,369\,750.342$

$c \approx \sqrt{11\,369\,750.342}$

6. Round off the answer to 2 decimal places. ≈ 3371.91

7. Write the answer. The rowers are 3371.91 m apart.

DISCUSSION

In what situations would you use the sine rule rather than the cosine rule?

Resources

eWorkbook Topic 15 Workbook (worksheets, code puzzle and project) (ewbk-13308)

Interactivity The cosine rule (int-6276)

Exercise 15.3 The cosine rule

learnon

15.3 Quick quiz on | 15.3 Exercise

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 2, 3, 4, 7, 9, 14, 17	5, 6, 8, 10, 15, 18	11, 12, 13, 16, 19

Where appropriate in this exercise, write your angles correct to the nearest minute and side lengths correct to 2 decimal places.

Fluency

1. WE4 Calculate the third side of triangle ABC given $a = 3.4$, $b = 7.8$ and $C = 80°$.
2. In triangle ABC, $b = 64.5$, $c = 38.1$ and $A = 58°34'$. Calculate the value of a.
3. In triangle ABC, $a = 17$, $c = 10$ and $B = 115°$. Calculate the value of b, and hence calculate the values of A and C.
4. WE5 Calculate the size of the smallest angle in the triangle with sides 6 cm, 4 cm and 8 cm. (*Hint:* The smallest angle is opposite the smallest side.)
5. In triangle ABC, $a = 356$, $b = 207$ and $c = 296$. Calculate the size of the largest angle.
6. In triangle ABC, $a = 23.6$, $b = 17.3$ and $c = 26.4$. Calculate the size of all the angles.
7. WE6 Two rowers set out from the same point. One rows N30°E for 1500 m and the other rows S40°E for 1200 m. Calculate the distance between the two rowers, correct to the nearest metre.

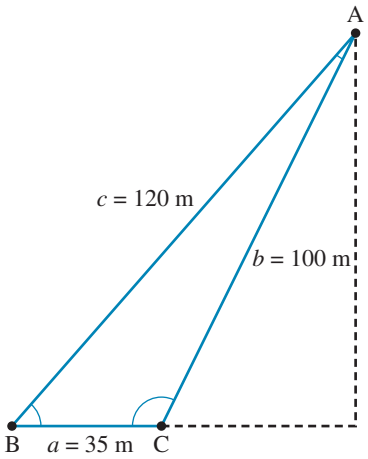
8. Maria cycles 12 km in a direction N68°W and then 7 km in a direction of N34°E.
   a. Calculate her distance from the starting point.
   b. Determine the bearing of the starting point from her finishing point.

Understanding

9. A garden bed is in the shape of a triangle with sides of length 3 m, 4.5 m and 5.2 m.
   a. Calculate the size of the smallest angle.
   b. Hence, calculate the area of the garden, correct to 2 decimal places.
   (*Hint:* Draw a diagram with the longest length as the base of the triangle.)

10. A hockey goal is 3 m wide. When Sophie is 7 m from one post and 5.2 m from the other, she shoots for goal. Determine within what angle, to the nearest degree, the shot must be made if it is to score a goal.

11. An advertising balloon is attached to two ropes 120 m and 100 m long. The ropes are anchored to level ground 35 m apart. Calculate the height of the balloon when both ropes are taut.



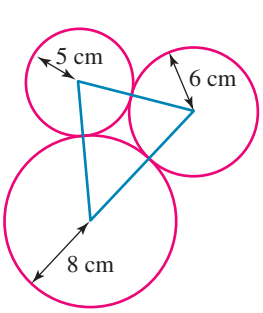
12. A plane flies in a direction of N70°E for 80 km and then on a bearing of S10°W for 150 km.
   a. Calculate the plane's distance from its starting point, correct to the nearest km.
   b. Calculate the plane's direction from its starting point.

13. Ship A is 16.2 km from port on a bearing of 053°T and ship B is 31.6 km from the same port on a bearing of 117°T.
   Calculate the distance between the two ships in km, correct to 1 decimal place.

Reasoning

14. A plane takes off at 10:00 am from an airfield and flies at 120 km/h on a bearing of N35°W. A second plane takes off at 10:05 am from the same airfield and flies on a bearing of S80°E at a speed of 90 km/h. Determine how far apart the planes are at 10:25 am in km, correct to 1 decimal place.

15. Three circles of radii 5 cm, 6 cm and 8 cm are positioned so that they just touch one another. Their centres form the vertices of a triangle.
   Determine the largest angle in the triangle. Show your working.

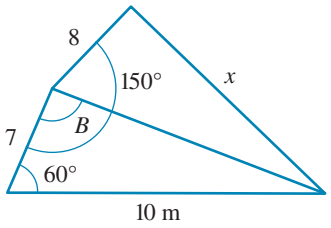


16. For the shape shown, determine:

a. the length of the diagonal

b. the magnitude (size) of angle B

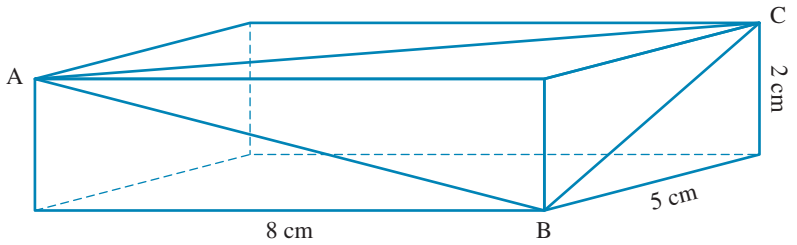
c. the length of x.



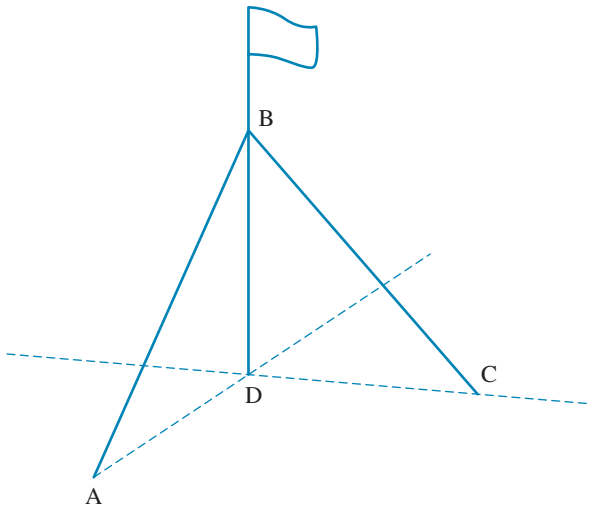
Problem solving

17. From the top of a vertical cliff 68 m high, an observer notices a yacht at sea. The angle of depression to the yacht is 47°. The yacht sails directly away from the cliff, and after 10 minutes the angle of depression is 15°. Determine the speed of the yacht in km/h, correct to 2 decimal places.

18. Determine the size of angles CAB, ABC and BCA.
Give your answers in degrees correct to 2 decimal places.



19. A vertical flag pole DB is supported by two wires AB and BC. AB is 5.2 metres long, BC is 4.7 metres long and B is 3.7 metres above ground level. Angle ADC is a right angle.



a. Evaluate the distance from A to C in metres, correct to 4 decimal places.

b. Determine the angle between AB and BC in degrees, correct to 2 decimal places.

LESSON
15.4 Area of triangles

LEARNING INTENTIONS

At the end of this lesson you should be able to:
- calculate the area of a triangle, given two sides and the included angle
- calculate the area of a triangle, given the three sides.

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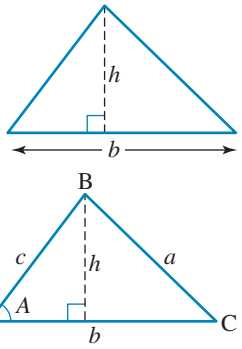
15.4.1 Areas of triangles when two sides and the included angle are known

- The area of any triangle is given by the rule area $= \frac{1}{2}bh$, where b is the base and h is the perpendicular height of the triangle.
- In the triangle ABC, b is the base and h is the perpendicular height of the triangle.
- Using the trigonometric ratio for sine:

$$\sin(A) = \frac{h}{c}$$

Transposing the equation to make h the subject, we obtain:

$$h = c\sin(A)$$



Area of a triangle

- **The area of triangle ABC can be found using the sine ratio:**

$$\text{Area} = \frac{1}{2}bc\sin(A)$$

- **Depending on how the triangle is labelled, the formula could read:**

$$\text{Area} = \frac{1}{2}ab\sin(C) \quad \text{Area} = \frac{1}{2}ac\sin(B) \quad \text{Area} = \frac{1}{2}bc\sin(A)$$

- The area formula may be used on any triangle provided that two sides of the triangle and the included angle (that is, the angle between the two given sides) are known.

WORKED EXAMPLE 7 Calculating the area of a triangle

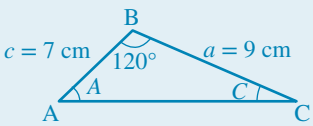
Calculate the area of the triangle shown.



THINK

1. Draw a labelled diagram of the triangle, label it ABC and fill in the given information.

WRITE/DRAW

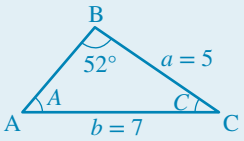


Let $a = 9$ cm, $c = 7$ cm, $B = 120°$.

2. Check that the criterion for the area rule has been satisfied.	The area rule can be used since two side lengths and the included angle have been given.
3. Write down the appropriate rule for the area.	$\text{Area} = \frac{1}{2}ac\sin(B)$
4. Substitute the known values into the rule.	$= \frac{1}{2} \times 9 \times 7 \times \sin 120°$
5. Evaluate. Round off the answer to 2 decimal places and include the appropriate unit.	$\approx 27.28\ \text{cm}^2$

WORKED EXAMPLE 8 Determining angles in a triangle and its area

A triangle has known dimensions of $a = 5$ cm, $b = 7$ cm and $B = 52°$. Determine A and C and hence the area.

THINK	WRITE/DRAW
1. Draw a labelled diagram of the triangle, label it ABC and fill in the given information.	 Let $a = 5$, $b = 7$, $B = 52°$.
2. Check whether the criterion for the area rule has been satisfied.	The area rule cannot be used since the included angle has not been given.
3. Write down the sine rule to calculate A.	To calculate angle A: $\frac{a}{\sin(A)} = \frac{b}{\sin(B)}$
4. Substitute the known values into the rule.	$\frac{5}{\sin(A)} = \frac{7}{\sin(52°)}$
5. Transpose the equation to make $\sin(A)$ the subject.	$5\sin(52°) = 7\sin(A)$ $\sin(A) = \frac{5\sin(52°)}{7}$
6. Evaluate.	$A = \sin^{-1}\left(\frac{5\sin(52°)}{7}\right)$ $\approx 34.254\,151\,87°$
7. Round off the answer to degrees and minutes.	$\approx 34°15'$
8. Determine the value of the included angle, C, using the fact that the angle sum of any triangle is 180°.	$C \approx 180° - (52° + 34°15')$ $= 93°45'$
9. Write down the appropriate rule for the area.	$\text{Area} = \frac{1}{2}ab\sin(C)$

10. Substitute the known values into the rule.

$$\approx \frac{1}{2} \times 5 \times 7 \times \sin(93°45')$$

11. Evaluate. Round off the answer to 2 decimal places and include the appropriate unit.

$$\approx 17.46\,\text{cm}^2$$

TI | THINK

In a new problem, open a Calculator page. To calculate the angle A, complete the entry line as:

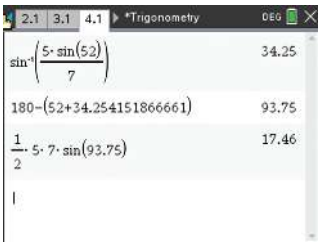
$$\sin^{-1}\left(\frac{5\ \sin(52)}{7}\right)$$

Note that you can leave the angle in decimal degrees and work with this value. Determine the value of C as shown in the screenshot. Then calculate the area by completing the entry line as:

$$\frac{1}{2} \times 5 \times 7 \sin(93.75)$$

Press ENTER after each entry.

DISPLAY/WRITE



$A = 34.25°$
$C = 93.75°$
The area of the triangle is $17.46\,\text{cm}^2$.

CASIO | THINK

To calculate the angle A, on the Main screen, complete the entry line as:

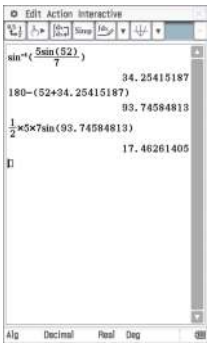
$$\sin^{-1}\left(\frac{5\ \sin(52)}{7}\right)$$

Note that we can leave the angle in decimal degrees and work with this value. Determine the value of C as shown. Then calculate the area by completing the entry line as:

$$\frac{1}{2} \times 5 \times 7 \sin(93.74584813)$$

Press EXE after each entry.

DISPLAY/WRITE



$A = 34.25°$
$C = 93.75°$
The area of the triangle is $17.46\,\text{cm}^2$.

eles-3216

15.4.2 Areas of triangles when three sides are known

- If the lengths of all the sides of the triangle are known but none of the angles are known, apply the cosine rule to calculate one of the angles.
- Substitute the values of the angle and the two adjacent sides into the area of a triangle formula Area $= \frac{1}{2}bc\ \sin(A)$.

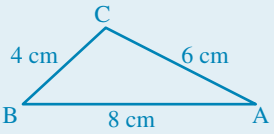
WORKED EXAMPLE 9 Calculating the area of a triangle given the lengths of the three sides

Calculate the area of the triangle with sides of 4cm, 6cm and 8cm.

THINK

1. Draw a labelled diagram of the triangle, call it ABC and fill in the given information.

WRITE/DRAW



Let $a = 4$, $b = 6$, $c = 8$.

2. Choose an angle and apply the cosine rule to calculate angle A.

$$\cos(A) = \frac{b^2 + c^2 - a^2}{2bc}$$
$$= \frac{6^2 + 8^2 - 4^2}{2 \times 6 \times 8}$$
$$= \frac{7}{8}$$
$$A = \cos^{-1}\left(\frac{7}{8}\right)$$
$$\approx 28.955\,024\,37°$$

3. Write down the appropriate rule for the area. $\text{Area} = \frac{1}{2}bc\sin(A)$

4. Substitute the known values into the rule. $\text{Area} = \frac{1}{2} \times 6 \times 8 \times \sin(28.955\,024\,37°)$

5. Evaluate. Round off the answer to 2 decimal places and include the appropriate unit. $\approx 11.618\,95$
The area is approximately $11.62\,\text{cm}^2$.

DISCUSSION

List three formulas for calculating the area of a triangle.

Resources

eWorkbook Topic 15 Workbook (worksheets, code puzzle and project) (ewbk-13308)
Interactivity Area of triangles (int-6483)

Exercise 15.4 Area of triangles

learnon

15.4 Quick quiz on | 15.4 Exercise

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 4, 7, 10, 12, 15, 18, 21	2, 5, 8, 11, 13, 16, 19, 22	3, 6, 9, 14, 17, 20, 23, 24

Where appropriate in this exercise, write your angles correct to the nearest minute and other measurements correct to 2 decimal places.

Fluency

1. WE7 Calculate the area of the triangle ABC with $a = 7$, $b = 4$ and $C = 68°$.
2. Calculate the area of the triangle ABC with $a = 7.3$, $c = 10.8$ and $B = 104°40'$.
3. Calculate the area of the triangle ABC with $b = 23.1$, $c = 18.6$ and $A = 82°17'$.
4. WE8 A triangle has $a = 10$ cm, $c = 14$ cm and $C = 48°$. Determine A and B and hence the area.
5. A triangle has $a = 17$ m, $c = 22$ m and $C = 56°$. Determine A and B and hence the area.
6. A triangle has $b = 32$ mm, $c = 15$ mm and $B = 38°$. Determine A and C and hence the area.
7. MC In a triangle, $a = 15$ m, $b = 20$ m and $B = 50°$. The area of the triangle is:
   A. $86.2\,\text{m}^2$ B. $114.9\,\text{m}^2$ C. $149.4\,\text{m}^2$ D. $172.4\,\text{m}^2$
8. WE9 Calculate the area of the triangle with sides of 5 cm, 6 cm and 8 cm.
9. Calculate the area of the triangle with sides of 40 mm, 30 mm and 5.7 cm.

10. Calculate the area of the triangle with sides of 16 mm, 3 cm and 2.7 cm.

11. MC A triangle has sides of length 10 cm, 14 cm and 20 cm. The area of the triangle is:

A. 41 cm^2 **B.** 65 cm^2 **C.** 106 cm^2 **D.** 137 cm^2

Understanding

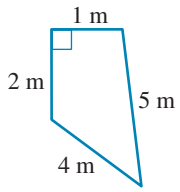
12. A piece of metal is in the shape of a triangle with sides of length 114 mm, 72 mm and 87 mm. Calculate its area.

13. A triangle has the largest angle of 115°. The longest side is 62 cm and another side is 35 cm. Calculate the area of the triangle to the nearest whole number.

14. A triangle has two sides of 25 cm and 30 cm. The angle between the two sides is 30°. Determine:

a. its area **b.** the length of its third side

15. The surface of a fish pond has the shape shown in the diagram.



Calculate how many goldfish the pond can support if each fish requires 0.3 m^2 surface area of water.

16. MC A parallelogram has sides of 14 cm and 18 cm and an angle between them of 72°. The area of the parallelogram is:

A. 118.4 cm^2 **B.** 172.4 cm^2 **C.** 239.7 cm^2 **D.** 252 cm^2

17. MC An advertising hoarding is in the shape of an isosceles triangle with sides of length 15 m, 15 m and 18 m. It is to be painted with two coats of purple paint. If the paint covers 12 m^2 per litre, the amount of paint needed, to the nearest litre, would be:

A. 9 L **B.** 18 L **C.** 24 L **D.** 36 L

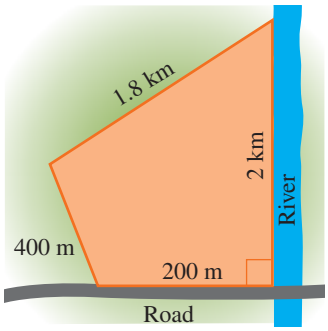
Reasoning

18. A parallelogram has diagonals of length 10 cm and 17 cm. An angle between them is 125°. Determine:

a. the area of the parallelogram **b.** the dimensions of the parallelogram.

19. A lawn is to be made in the shape of a triangle with sides of length 11 m, 15 m and 17.2 m. Determine how much grass seed, to the nearest kilogram, needs to be purchased if it is sown at the rate of 1 kg per 5 m^2.

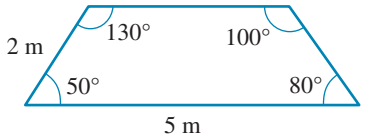
20. A bushfire burns out an area of level grassland shown in the diagram.



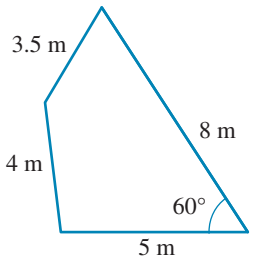
(*Note:* This is a sketch of the area and is not drawn to scale.)
Evaluate the area, in hectares correct to 1 decimal place, of the land that is burned.

Problem solving

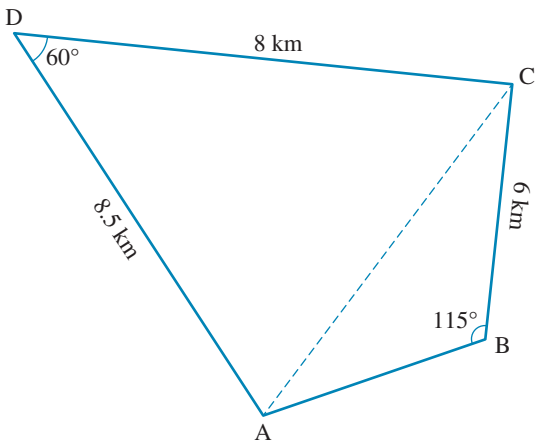
21. An earth embankment is 27 m long and has a vertical cross-section as shown in the diagram. Determine the volume of earth needed to build the embankment, correct to the nearest cubic metre.



22. Evaluate the area of this quadrilateral.

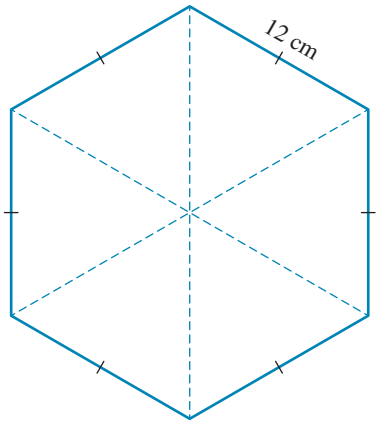


23. A surveyor measured the boundaries of a property as shown. The side AB could not be measured because it crossed through a marsh.
The owner of the property wanted to know the total area and the length of the side AB.
Give all lengths correct to 2 decimal places and angles to the nearest degree.



a. Calculate the area of the triangle ACD.
b. Calculate the distance AC.
c. Calculate the angle CAB.
d. Calculate the angle ACB.
e. Calculate the length AB.
f. Determine the area of the triangle ABC.
g. Determine the area of the property.

24. A regular hexagon has sides of length 12 centimetres. It is divided into six smaller equilateral triangles. Evaluate the area of the hexagon, giving your answer correct to 2 decimal places.



LESSON
15.5 The unit circle

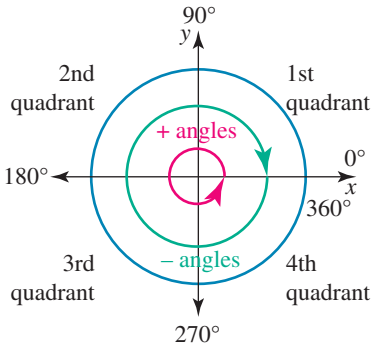
LEARNING INTENTIONS

At the end of this lesson you should be able to:
- determine in which quadrant an angle lies
- use and interpret the relationship between a point on the unit circle and the angle made with the positive x-axis
- use the unit circle to determine approximate trigonometric ratios for angles greater than 90°
- use and interpret the relationships between supplementary and complementary angles.

eles-5009

15.5.1 The unit circle

- A **unit circle** is a circle with a radius of 1 unit.
- The unit circle is divided into 4 quadrants, numbered in an anticlockwise direction, as shown in the diagram.
- *Positive angles* are measured anticlockwise from 0°.
- *Negative angles* are measured clockwise from 0°.



WORKED EXAMPLE 10 Identifying where an angle lies on the unit circle

State the quadrant of the unit circle in which each of the following angles is found.
a. **145°** **b.** **282°**

THINK

a. The given angle is between 90° and 180°. State the appropriate quadrant.

b. The given angle is between 270° and 360°. State the appropriate quadrant.

WRITE

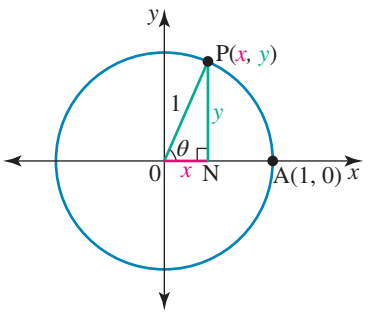
a. 145° is in quadrant 2.

b. 282° is in quadrant 4.

- Consider the unit circle with point P(x, y) making the right-angled triangle OPN as shown in the diagram.
- Using the trigonometric ratios:

$$\frac{x}{1} = \cos(\theta), \quad \frac{y}{1} = \sin(\theta), \quad \frac{y}{x} = \tan(\theta)$$

where θ is measured anticlockwise from the positive x-axis.



Calculating values of sine, cosine and tangent

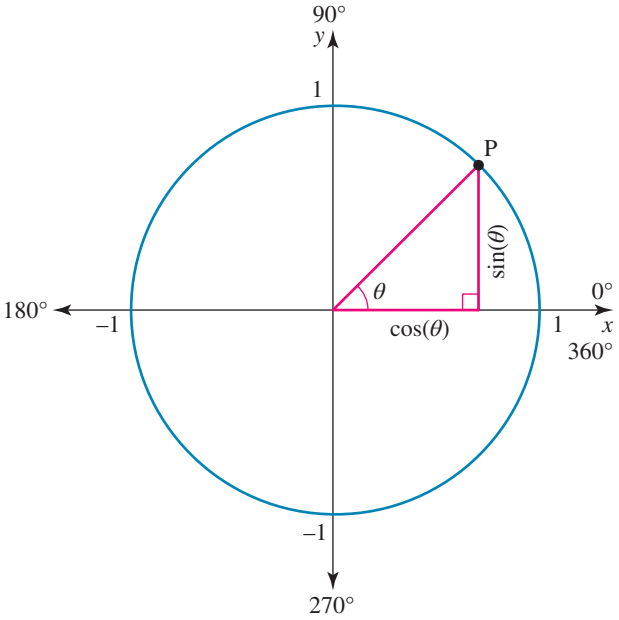
To calculate the value of sine, cosine or tangent of any angle θ from the unit circle:

$$\cos(\theta) = x$$
$$\sin(\theta) = y$$
$$\tan(\theta) = \frac{y}{x} = \frac{\sin(\theta)}{\cos(\theta)}$$

eles-5010

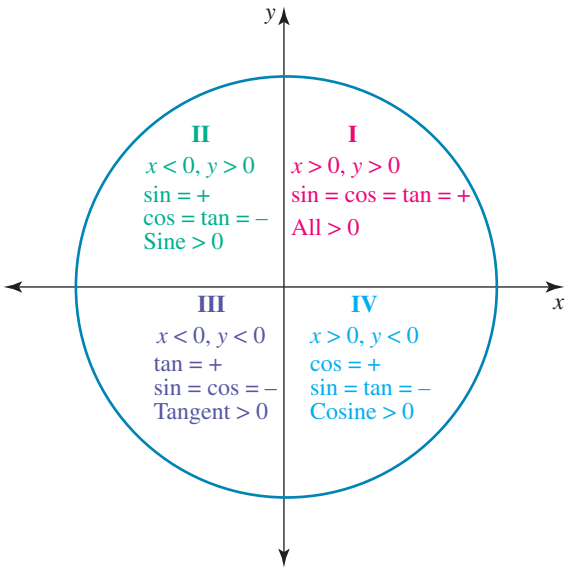
15.5.2 The four quadrants of the unit circle

- Approximate values for sine, cosine and tangent of an angle can be found from the unit circle using the following steps, as shown in the diagram.
  Step 1: Draw a unit circle and label the x- and y-axes.
  Step 2: Mark the angles 0°, 90°, 180°, 270° and 360°.
  Step 3: Draw the given angle θ.
  Step 4: Mark $x = \cos(\theta)$, $y = \sin(\theta)$.
  Step 5: Approximate the values of x and y and equate to give the values of $\cos(\theta)$ and $\sin(\theta)$.
- Where the angle lies in the unit circle determines whether the trigonometric ratio is positive or negative.



Sign of the trigonometric functions

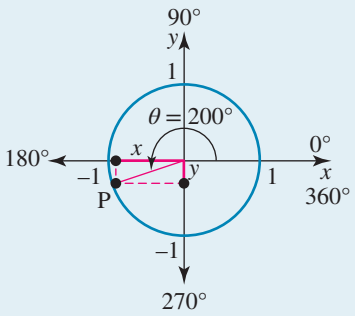
- Consider the following.
  - In the first quadrant $x > 0$, $y > 0$; therefore, **All** trig ratios are positive.
  - In the second quadrant $x < 0$, $y > 0$; therefore, **Sine** (the y-value) is positive.
  - In the third quadrant $x < 0$, $y < 0$; therefore, **Tangent** $\left(\frac{y}{x} \text{ values}\right)$ is positive.
  - In the fourth quadrant $x > 0$, $y < 0$; therefore, **Cosine** (the x-value) is positive.
  - There are several mnemonics for remembering the sign of the trigonometric functions.
    A common saying is '**A**ll **S**tations **T**o **C**entral'.



WORKED EXAMPLE 11 Using the unit circle to approximate the trigonometric ratios of an angle

Determine the approximate value of each of the following using the unit circle.

a. **sin(200°)** **b.** **cos(200°)** **c.** **tan(200°)**

THINK	WRITE/DRAW
Draw a unit circle and construct an angle of 200°. Label the point corresponding to the angle of 200° on the circle P. Highlight the lengths representing the x- and y-coordinates of point P.	
a. The sine of the angle is given by the y-coordinate of P. Determine the y-coordinate of P by measuring the distance along the y-axis. State the value of sin(200°). (*Note:* The sine value will be negative as the y-coordinate is negative.)	**a.** $\sin(200°) = -0.3$
b. The cosine of the angle is given by the x-coordinate of P. Determine the x-coordinate of P by measuring the distance along the x-axis. State the value of cos(200°). (*Note:* Cosine is also negative in quadrant 3, as the x-coordinate is negative.)	**b.** $\cos(200°) = -0.9$
c. $\tan(200°) = \dfrac{\sin(200°)}{\cos(200°)}$	**c.** $\dfrac{-0.3}{-0.9} = \dfrac{1}{3} = 0.3333$

- The approximate results obtained in Worked example 11 can be verified with the aid of a calculator:

  $\sin(200°) = -0.342\,020\,143$, $\cos(200°) = -0.939\,692\,62$ and $\tan(200°) = 0.3640$.

  Rounding these values to 1 decimal place would give -0.3, -0.9 and 0.4 respectively, which match the values obtained from the unit circle.

eles-3217

15.5.3 The relationship between supplementary and complementary angles

Supplementary angles

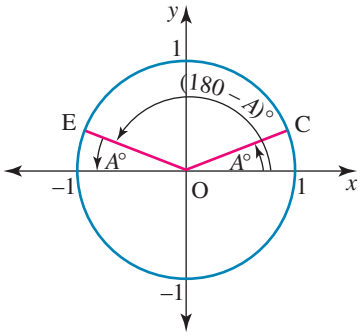
- Consider the special relationship between the sine, cosine and tangent of supplementary angles, say $A°$ and $(180 - A)°$.

  In the diagram, the y-axis is an axis of symmetry.
  - The y-values of points C and E are the same.
    That is, $\sin(A°) = \sin(180 - A)°$.
  - The x-values of points C and E are opposites in value.
    That is, $\cos(A°) = -\cos(180 - A)°$

  Thus: $\sin(180 - A)° = \sin(A°)$

  $\cos(180 - A)° = -\cos(A°)$

  $$\tan(180 - A)° = \frac{\sin(180 - A)°}{\cos(180 - A)°} = \frac{\sin(A°)}{-\cos(A°)} = -\tan(A°).$$



Supplementary angles $A°$ and $(180° - A)$ where $0° \leq A \leq 90°$

$$\sin(180° - A) = \sin(A)$$

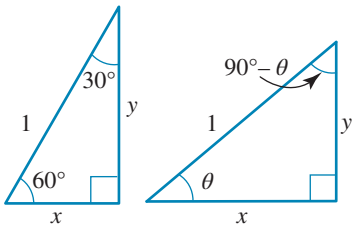
$$\cos(180° - A) = -\cos(A)$$

$$\tan(180° - A) = -\tan(A)$$

Complementary angles

- Complementary angles add to 90°. Therefore, 30° and 60° are complementary angles, and θ and $(90° - \theta)$ are also complementary angles.
- The sine of an angle is equal to the cosine of its complement. Therefore, $\sin(60°) = \cos(30°)$.
- We say that sine and cosine are complementary functions.

$$\sin A = \cos(90° - A)$$
$$\cos A = \sin(90° - A)$$



Complementary angles where $0° \leq A \leq 90°$

$$\sin A = \cos(90° - A)$$

$$\cos A = \sin(90° - A)$$

on Resources

eWorkbook Topic 15 Workbook (worksheets, code puzzle and project) (ewbk-13308)

Exercise 15.5 The unit circle

learnon

15.5 Quick quiz on | **15.5 Exercise**

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 2, 6, 8, 12, 17, 20	3, 4, 7, 9, 11, 13, 18, 21	5, 10, 14, 15, 16, 19, 22

Where appropriate in this exercise, give answers correct to 2 decimal places.

Fluency

1. WE10 State which quadrant of the unit circle each of the following angles is in.

a. 60° b. 130° c. 310° d. 260° e. 100° f. 185°

2. MC If $\theta = 43°$, the triangle drawn to show this would be in:

   A. quadrant 1 B. quadrant 2 C. quadrant 3 D. quadrant 4

3. MC If $\theta = 295°$, the triangle drawn to show this would be in:

   A. quadrant 1 B. quadrant 2 C. quadrant 3 D. quadrant 4

4. WE11 Determine the approximate value of each of the following using the unit circle.

   a. sin(20°) b. cos(20°) c. cos(100°) d. sin(100°)

5. Determine the approximate value of each of the following using the unit circle.

   a. sin(320°) b. cos(320°) c. sin(215°) d. cos(215°)

6. Use the unit circle to determine the approximate value of each of the following.

   a. sin(90°) b. cos(90°) c. sin(180°) d. cos(180°)

7. Use the unit circle to determine the approximate value of each of the following.

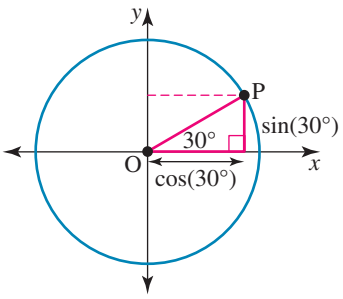
   a. sin(270°) b. cos(270°) c. sin(360°) d. cos(360°)

Understanding

8. On the unit circle, use a protractor to measure an angle of 30° from the positive x-axis. Mark the point P on the circle. Use this point to construct a triangle in quadrant 1 as shown.

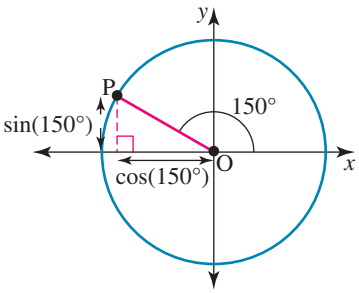
   a. Calculate the value of cos(30°).
   (Remember that the length of the adjacent side of the triangle is cos(30°).)
   b. Calculate the value of sin(30°).
   (This is the length of the opposite side of the triangle.)

   Check your answers in a and b by calculating these values with a calculator.



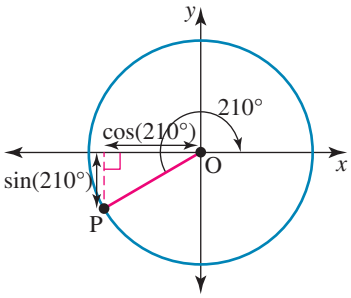
9. Using a graph of the unit circle, measure 150° with a protractor and mark the point P on the circle. Use this point to draw a triangle in quadrant 2 as shown.

   a. Determine the angle the radius OP makes with the negative x-axis.
   b. Remembering that $x = \cos(\theta)$, use your circle to determine the value of cos(150°).
   c. Comment on how cos(150°) compares to cos(30°).
   d. Remembering that $y = \sin(\theta)$, use your circle to determine the value of sin(150°).
   e. Comment on how sin(150°) compares with sin(30°).

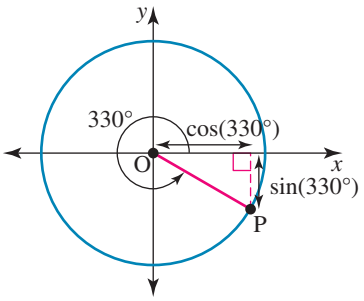


10. On a unit circle, measure 210° with a protractor and mark the point P on the circle.
    Use this point to draw a triangle in quadrant 3 as shown.

    a. Determine the angle the radius OP makes with the negative x-axis.
    b. Use your circle to determine the value of cos(210°).
    c. Comment on how cos(210°) compares to cos(30°).
    d. Use your circle to determine the value of sin(210°).
    e. Comment on how sin(210°) compares with sin(30°).

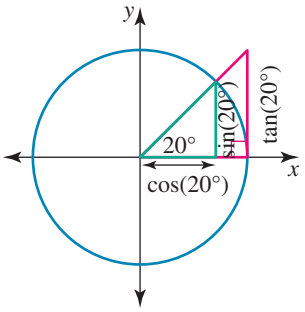


11. On a unit circle, measure 330° with a protractor and mark the point P on the circle. Use this point to draw a triangle in quadrant 4 as shown.



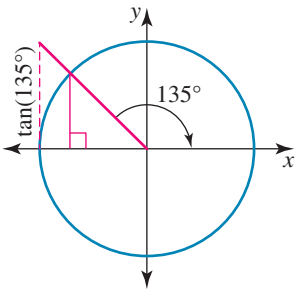
a. Determine the angle the radius OP makes with the positive x-axis.
b. Use your circle to determine the value of cos(330°).
c. Comment on how cos(330°) compares to cos(30°).
d. Use your circle to determine the value of sin(330°).
e. Comment on how sin(330°) compares with sin(30°).

12. On a unit circle, draw an appropriate triangle for the angle of 20° in quadrant 1.



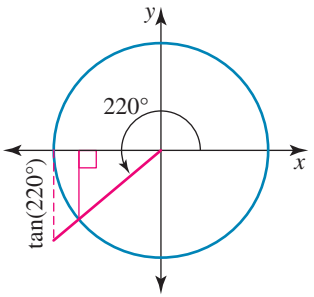
a. Determine the value of sin(20°).
b. Determine the value of cos(20°).
c. Draw a tangent line and extend the hypotenuse of the triangle to meet the tangent as shown.
Accurately measure the length of the tangent between the x-axis and the point where it meets the hypotenuse, and hence state the value of tan(20°).
d. Determine the value of $\frac{\sin(20°)}{\cos(20°)}$.
e. Comment on how tan(20°) compares with $\frac{\sin(20°)}{\cos(20°)}$.

13. On a unit circle, draw an appropriate triangle for the angle of 135° in quadrant 2.



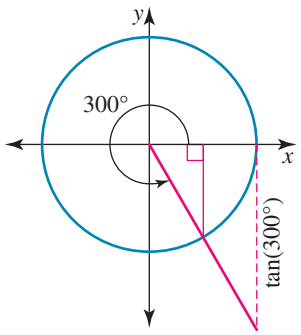
a. Determine the value of sin(135°) using sin(45°).
b. Determine the value of cos(135°) using cos(45°).
c. Draw a tangent line and extend the hypotenuse of the triangle to meet the tangent as shown.
Accurately measure the length of the tangent to where it meets the hypotenuse to calculate the value of tan(135°).
d. Determine the value of $\frac{\sin(135°)}{\cos(135°)}$.
e. Comment on how tan(135°) compares with $\frac{\sin(135°)}{\cos(135°)}$.
f. Comment on how tan(135°) compares with tan(45°).

14. On a unit circle, draw an appropriate triangle for the angle of 220° in quadrant 3.



a. Determine the value of sin(220°).
b. Determine the value of cos(220°).
c. Draw a tangent line and extend the hypotenuse of the triangle to meet the tangent as shown.
Determine the value of tan(220°) by accurately measuring the length of the tangent to where it meets the hypotenuse.
d. Determine the value of $\frac{\sin(220°)}{\cos(220°)}$.
e. Comment on how tan(220°) compares with $\frac{\sin(220°)}{\cos(220°)}$.
f. Comment on how tan(220°) compares with tan(40°). (Use a calculator.)

15. On a unit circle, draw an appropriate triangle for the angle of 300° in quadrant 4.



a. Determine the value of $\sin(300°)$.
b. Determine the value of $\cos(300°)$.
c. Draw a tangent line and extend the hypotenuse of the triangle to meet the tangent as shown. Determine the value of $\tan(300°)$ by accurately measuring the length of the tangent to where it meets the hypotenuse.
d. Determine the value of the value of $\dfrac{\sin(300°)}{\cos(300°)}$.
e. Comment on how $\tan(300°)$ compares with $\dfrac{\sin(300°)}{\cos(300°)}$.
f. Comment on how $\tan(300°)$ compares with $\tan(60°)$. (Use a calculator.)

16. MC In a unit circle, the length of the radius is equal to:

A. $\sin(\theta)$ **B.** $\cos(\theta)$ **C.** $\tan(\theta)$ **D.** 1

Reasoning

17. Show that $\sin^2(\alpha°) + \cos^2(\alpha°) = 1$.

18. Show that $1 - \sin^2(180 - \alpha)° = \cos^2(180 - \alpha)°$.

19. Show that $1 + \tan^2(\alpha°) = \sec^2(\alpha°)$, where $\sec(\alpha°) = \dfrac{1}{\cos(\alpha°)}$.

Problem solving

20. If $\sin(x°) = p$, $0 \le x \le 90°$, write each of the following in terms of p.
a. $\cos(x°)$
b. $\sin(180 - x)°$
c. $\cos(180 - x)°$
d. $\cos(90 - x)°$

21. Simplify $\sin(180 - x)° - \sin(x°)$.

22. Simplify $\cos(180 - x)° + \cos(x°)$, where $0 < x° < 90°$.

LESSON
15.6 Trigonometric functions

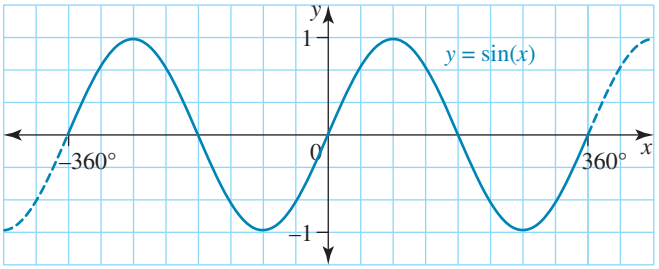
LEARNING INTENTIONS

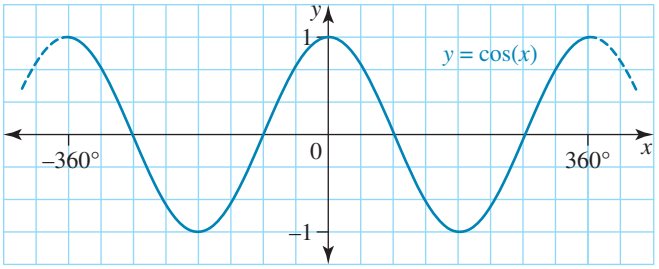
At the end of this lesson you should be able to:
- sketch the graphs of the sine, cosine and tangent functions
- determine the amplitude of a given trigonometric function
- determine the period of a given trigonometric function.

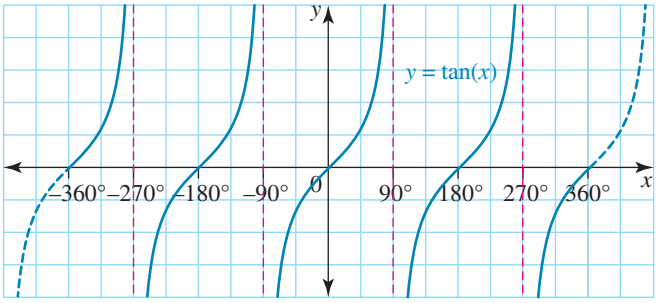
15.6.1 Sine, cosine and tangent graphs

eles-5011

- The graphs of $y = \sin(x)$, $y = \cos(x)$ and $y = \tan(x)$ are shown below.







- Trigonometric graphs repeat themselves continuously in cycles; hence, they are called **periodic functions**.
- The **period** of the graph is the horizontal distance between repeating peaks or troughs. The period between the repeating peaks for $y = \sin(x)$ and $y = \cos(x)$ is 360°. The period of the graph of $y = \tan(x)$ is 180°, and **asymptotes** occur at $x = 90°$ and intervals of 180°.
- The **amplitude** of a periodic graph is half the distance between the maximum and minimum values of the function.
  Amplitude can also be described as the amount by which the graph goes above and below its mean value, which is the x-axis for the graphs of $y = \sin(x)$, $y = \cos(x)$ and $y = \tan(x)$.

- The following can be summarised from the graphs of the trigonometric functions.

Graph	Period	Amplitude
$y = \sin(x)$	360°	1
$y = \cos(x)$	360°	1
$y = \tan(x)$	180°	Undefined

- A scientific calculator can be used to check significant points such as 0°, 90°, 180°, 270° and 360° for the shape of a sin, cos or tan graph.

Transformations of trigonometric graphs

- **The sine, cosine and tangent graphs can be dilated, translated and reflected in the same way as other functions.**
- **These transformations are summarised in the table below.**

Graph	Period	Amplitude
$y = a\sin(nx)$	$\frac{360°}{n}$	a
$y = a\cos(nx)$	$\frac{360°}{n}$	a
$y = a\tan(nx)$	$\frac{180°}{n}$	Undefined

- **If $a < 0$, the graph is reflected in the x-axis. The amplitude is always the positive value of a.**

WORKED EXAMPLE 12 Sketching periodic functions

Sketch the graphs of a $y = 2\sin(x)$ and b $y = \cos(2x)$ for $0° \leq x \leq 360°$.

THINK

a. 1. The graph must be drawn from 0° to 360°.

2. Compared to the graph of $y = \sin(x)$ each value of $\sin(x)$ has been multiplied by 2; therefore, the amplitude of the graph must be 2.

3. Label the graph $y = 2\sin(x)$.

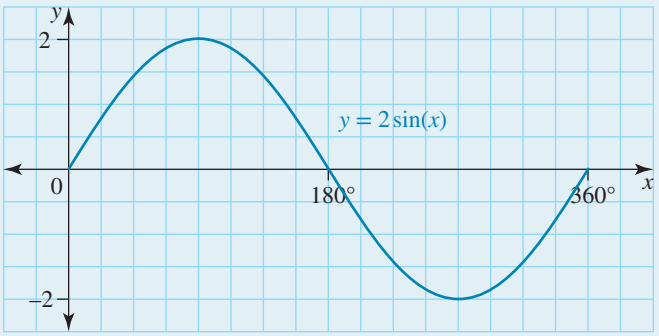
b. 1. The graph must be drawn from 0° to 360°.

2. Compared to the graph of $y = \cos(x)$, each value of x has been multiplied by 2; therefore, the period of the graph must become 180°.

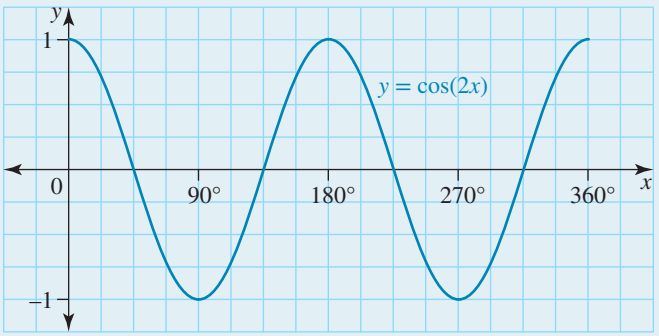
3. Label the graph $y = \cos(2x)$.

WRITE/DRAW

a.



b.



TI | THINK

a.

1. In a new problem, on a Graphs page, ensure the Graphs & Geometry Settings are set to the degrees mode, as shown in the screenshot.
   To do this, press:
   - MENU
   - 9: Settings

   For Display Digits select 'Fix 2'.
   TAB to Graphing Angle and select 'Degree'. TAB to OK and press ENTER.
2. To set an appropriate viewing window, press:
   - MENU
   - 4: Window/Zoom
   - 1: Window Settings...

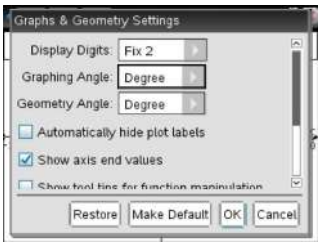
   Select the values as shown in the screenshot.
   TAB to OK and press ENTER.
3. Complete the function entry line as:
   $f1(x) = 2\sin(x) \mid 0 \le x \le 360$
   Press ENTER. The graph is displayed as required for $0° \le x \le 360°$.

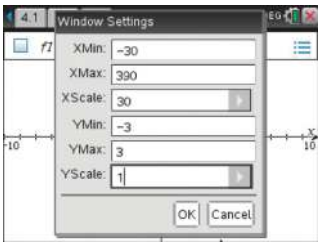
b.

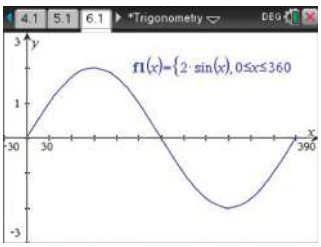
Complete the function entry line as:
$f2(x) = \cos(2x) \mid 0 \le x \le 360$
Press ENTER, and the graph is displayed, as required only for $0° \le x \le 360°$.

DISPLAY/WRITE

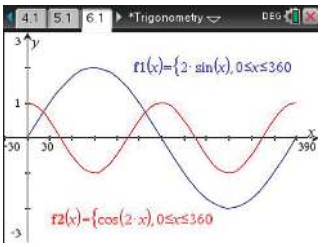
a.







b.



CASIO | THINK

On a Graph & Table screen, set an appropriate viewing window as shown.

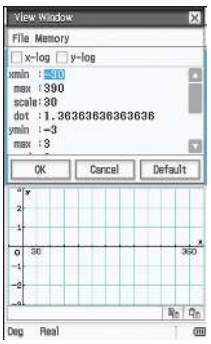
a.

Complete the function entry line as:
$y1 = 2\sin(x) \mid 0 \le x \le 360$
Press EXE.
Tap the graphing icon and the graph is displayed as required only for $0° \le x \le 360°$.

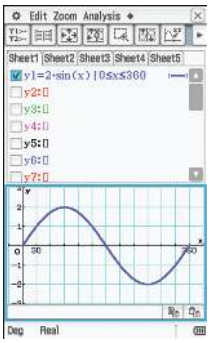
b.

Complete the function entry line as:
$y1 = \cos(2x) \mid 0 \le x \le 360$
Press EXE.
Tap the graphing icon and the graph is displayed as required only for $0° \le x \le 360°$.

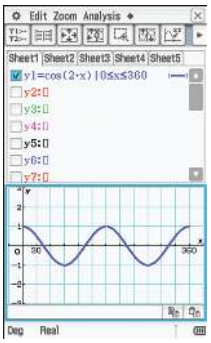
DISPLAY/WRITE



a.



b.



WORKED EXAMPLE 13 Stating the amplitude and period of periodic functions

For each of the following graphs, state:
i. the amplitude
ii. the period.

a. $y = 2\sin(3x)$

b. $y = \cos\left(\frac{x}{3}\right)$

c. $y = \tan(2x)$

THINK	WRITE
a. The value of a is 2.	**a. i.** Amplitude $= 2$
The period is $\frac{360°}{n}$.	**ii.** Period $= \frac{360}{3} = 120°$
b. The value of a is 1.	**b. i.** Amplitude $= 1$
The period is $\frac{360°}{n}$.	**ii.** Period $= \frac{360}{\frac{1}{3}} = 1080°$
c. The tangent curve has an undefined amplitude.	**c. i.** Amplitude $=$ undefined
The period is $\frac{180°}{2}$.	**ii.** Period $= \frac{180°}{2} = 90°$

DISCUSSION

For the graph of $y = a\tan(nx)$, what would be the period and amplitude?

on Resources

eWorkbook Topic 15 Workbook (worksheets, code puzzle and project) (ewbk-13308)
Interactivity Graphs of trigonometric functions (int-4821)

Exercise 15.6 Trigonometric functions

learnon

15.6 Quick quiz on | 15.6 Exercise

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 4, 8, 12, 16, 20, 25, 26, 28, 29, 32	3, 5, 7, 10, 13, 14, 17, 21, 23, 27, 30, 33	2, 6, 9, 11, 15, 18, 19, 22, 24, 31, 34

Fluency

1. Using your calculator (or the unit circle if you wish), complete the following table.

x	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\sin(x)$													
x	390°	420°	450°	480°	510°	540°	570°	600°	630°	660°	690°	720°	
$\sin(x)$													

For questions 2 to 7, using graph paper, rule x- and y-axes and carefully mark a scale along each axis.

2. Use 1 cm = 30° on the x-axis to show x-values from 0° to 720°.
   Use 2 cm = 1 unit along the y-axis to show y-values from −1 to 1.
   Carefully plot the graph of $y = \sin(x)$ using the values from the table in question 1.

3. State how long it takes for the graph of $y = \sin(x)$ to complete one full cycle.

4. From your graph of $y = \sin(x)$, estimate to 1 decimal place the value of y for each of the following.
   a. $x = 42°$ b. $x = 130°$ c. $x = 160°$ d. $x = 200°$

5. From your graph of $y = \sin(x)$, estimate to 1 decimal place the value of y for each of the following.
   a. $x = 180$ b. $x = 70°$ c. $x = 350°$ d. $x = 290°$

6. From your graph of $y = \sin(x)$, estimate to the nearest degree a value of x for each of the following.
   a. $y = 0.9$ b. $y = -0.9$ c. $y = 0.7$

7. From your graph of $y = \sin(x)$, estimate to the nearest degree a value of x for each of the following.
   a. $y = -0.5$ b. $y = -0.8$ c. $y = 0.4$

8. Using your calculator (or the unit circle if you wish), complete the following table.

x	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\cos(x)$													
x	390°	420°	450°	480°	510°	540°	570°	600°	630°	660°	690°	720°	
$\cos(x)$													

For questions 9 to 14, using graph paper, rule x- and y-axes and carefully mark a scale along each axis.

9. Use 1 cm = 30° on the x-axis to show x-values from 0° to 720°.
   Use 2 cm = 1 unit along the y-axis to show y-values from −1 to 1.
   Carefully plot the graph of $y = \cos(x)$ using the values from the table in question 8.

10. If you were to continue the graph of $y = \cos(x)$, state what shape you would expect it to take.

11. State whether the graph of $y = \cos(x)$ is the same as the graph of $y = \sin(x)$.
    Explain how it differs. State what features are the same.

12. Using the graph of $y = \cos(x)$, estimate to 1 decimal place the value of y for each of the following.

a. $x = 48°$ **b.** $x = 155°$ **c.** $x = 180°$ **d.** $x = 340°$

13. Using the graph of $y = \cos(x)$, estimate to 1 decimal place the value of y for each of the following.

a. $x = 240°$ **b.** $x = 140°$ **c.** $x = 40°$ **d.** $x = 200°$

14. Using the graph of $y = \cos(x)$, estimate to the nearest degree a value of x for each of the following.

a. $y = -0.5$ **b.** $y = 0.8$ **c.** $y = 0.7$

15. Using the graph of $y = \cos(x)$, estimate to the nearest degree a value of x for each of the following.

a. $y = -0.6$ **b.** $y = 0.9$ **c.** $y = -0.9$

16. Using your calculator (or the unit circle if you wish), complete the following table.

x	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\tan(x)$													
x	390°	420°	450°	480°	510°	540°	570°	600°	630°	660°	690°	720°	
$\tan(x)$													

For questions **17** to **22**, using graph paper, rule x- and y-axes and carefully mark a scale along each axis.

17. Use 1 cm = 30° on the x-axis to show x-values from 0° to 720°.
Use 2 cm = 1 unit along the y-axis to show y-values from −2 to 2.
Carefully plot the graph of $y = \tan(x)$ using the values from the table in question **16**.

18. If you were to continue the graph of $y = \tan(x)$, state what shape would you expect it to take.

19. State whether the graph of $y = \tan(x)$ is the same as the graphs of $y = \sin(x)$ and $y = \cos(x)$.
Explain how it differs. State what features are the same.

20. Using the graph of $y = \tan(x)$, estimate to 1 decimal place the value of y for each of the following.

a. $x = 60°$ **b.** $x = 135°$ **c.** $x = 310°$ **d.** $x = 220°$

21. Using the graph of $y = \tan(x)$, determine the value of y for each of the following.

a. $x = 500°$ **b.** $x = 590°$ **c.** $x = 710°$ **d.** $x = 585°$

22. Using the graph of $y = \tan(x)$, estimate to the nearest degree a value of x for each of the following.

a. $y = 1$ **b.** $y = 1.5$ **c.** $y = -0.4$
d. $y = -2$ **e.** $y = 0.2$ **f.** $y = -1$

WE12&13 For each of the graphs in questions **23** and **24**:

i. state the period **ii.** state the amplitude **iii.** sketch the graph.

23. **a.** $y = \cos(x)$, for $x \in [-180°, 180°]$
b. $y = \sin(x)$, for $x \in [0°, 720°]$

24. **a.** $y = \sin(2x)$, for $x \in [0°, 360°]$
b. $y = 2\cos(x)$, for $x \in [-360°, 0°]$

25. For each of the following, state:

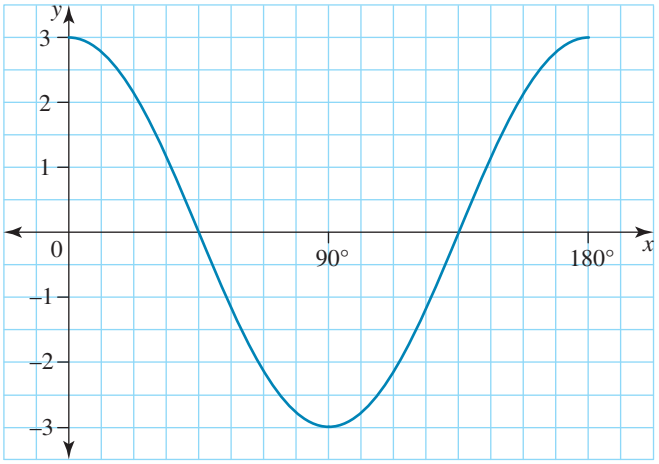
i. the period **ii.** the amplitude.

a. $y = 3\cos(2x)$ **b.** $y = 4\sin(3x)$ **c.** $y = 2\cos\left(\frac{x}{2}\right)$

d. $y = \frac{1}{2}\sin\left(\frac{x}{4}\right)$ **e.** $y = -\sin(x)$ **f.** $y = -\cos(2x)$

Understanding

26. MC Use the graph shown to answer the following.



a. The amplitude of the graph is:

A. 180°
B. 90°
C. 3
D. −3

b. The period of the graph is:

A. −3
B. 3
C. 90°
D. 360°

c. The equation of the graph could be:

A. $y = \cos(x)$
B. $y = \sin(x)$
C. $y = 3\cos\left(\frac{x}{3}\right)$
D. $y = 3\cos(2x)$

27. Sketch each of the following graphs, stating the period and amplitude of each.

a. $y = 2\cos\left(\frac{x}{3}\right)$, for $x \in [0°, 1080°]$

b. $y = -3\sin(2x)$, for $x \in [0°, 360°]$

c. $y = 3\sin\left(\frac{x}{2}\right)$, for $x \in [-180°, 180°]$

d. $y = -\cos(3x)$, for $x \in [0°, 360°]$

e. $y = 5\cos(2x)$, for $x \in [0°, 180°]$

f. $y = -\sin(4x)$, for $x \in [0°, 180°]$

28. Use technology to sketch the graphs of each of the following for $0° \leq x \leq 360°$.

a. $y = \cos(x) + 1$

b. $y = \sin(2x) - 2$

c. $y = \cos\left(\frac{\pi}{180}(x - 60)\right)$

d. $y = 2\sin(4x) + 3$

Reasoning

29. **a.** Sketch the graph of $y = \cos(2x)$ for $x \in [0°, 360°]$.
- **i.** State the minimum value of y for this graph.
- **ii.** State the maximum value of y for this graph.

b. Using the answers obtained in part **a**, write down the maximum and minimum values of $y = \cos(2x) + 2$.

c. Determine what would be the maximum and minimum values of the graph of $y = 2\sin(x) + 3$. Explain how you obtained these values.

30. **a.** Complete the table below by filling in the exact values of $y = \tan(x)$

x	0°	30°	60°	90°	120°	150°	180°
$y = \tan(x)$							

b. Sketch the graph of $y = \tan(x)$ for $[0°, 180°]$.

c. Determine what happens at $x = 90°$.

d. For the graph of $y = \tan(x)$, $x = 90°$ is called an asymptote. Write down when the next asymptote would occur.

e. Determine the period and amplitude of $y = \tan(x)$.

31. Answer the following questions.

a. Sketch the graph of $y = \tan(2x)$ for $[0°, 180°]$.

b. Determine when the asymptotes occur.

c. State the period and amplitude of $y = \tan(2x)$.

Problem solving

32. The height, h metres, of the tide above the mean sea level on the first day of the month is given by the rule

$$h = 3\sin(30t°)$$

where t is the time in hours since midnight.

a. Sketch the graph of h versus t.

b. Determine the height of the high tide.

c. Calculate the height of the tide at 8 am.

33. The height, h metres, of the tide on the first day of January at Trig Cove is given by the rule

$$h = 6 + 4\sin(30t°)$$

where t is the time in hours since midnight.

a. Sketch the graph of h versus t, for $0 \leq t \leq 24$.

b. Determine the height of the high tide.

c. Determine the height of the low tide.

d. Calculate the height of the tide at 10 am, correct to the nearest centimetre.

34. The temperature, T, inside a house t hours after 3 am is given by the rule

$$T = 22 - 2\cos(15t°) \text{ for } 0 \leq t \leq 24$$

where T is the temperature in degrees Celsius.

a. Determine the temperature inside the house at 9 am.

b. Sketch the graph of T versus t.

c. Determine the warmest and coolest temperatures that it gets inside the house over the 24-hour period.

LESSON
15.7 Solving trigonometric equations

LEARNING INTENTIONS

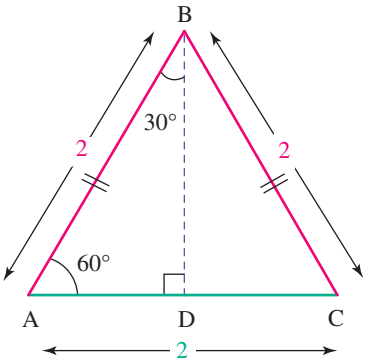
At the end of this lesson you should be able to:

- apply the exact values of sin, cos and tan for 30°, 45° and 60° angles
- solve trigonometric equations graphically for a given domain
- solve trigonometric equations algebraically, using exact values, for a given domain.

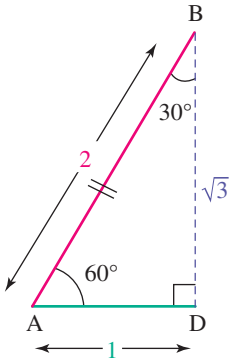
eles-3215

15.7.1 Exact values of trigonometric functions

- Most of the trigonometric values that we will deal with in this topic are approximations.
- However, angles of 30°, 45° and 60° have exact values of sine, cosine and tangent.
- Consider an equilateral triangle, ABC, of side length 2 cm.
  Let BD be the perpendicular bisector of AC, then:
  $\Delta ABD \cong \Delta CBD$ (using RHS)
  giving:
  $AD = CD = 1$ cm
  $\angle ABD = \angle CBD = 30°$
  and:
  $(AB)^2 = (AD)^2 + (BD)^2$ (using Pythagoras' theorem)
  $2^2 = 1^2 + (BD)^2$
  $BD = \sqrt{3}$



- Using ΔABD, the following exact values are obtained:



$$\sin(A) = \frac{\text{opp}}{\text{hyp}} \Rightarrow \sin(60°) = \frac{\sqrt{3}}{2}$$

$$\cos(A) = \frac{\text{adj}}{\text{hyp}} \Rightarrow \cos(60°) = \frac{1}{2}$$

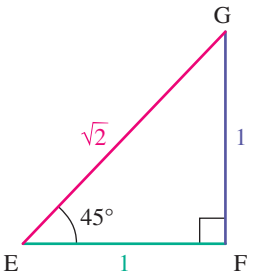
$$\tan(A) = \frac{\text{opp}}{\text{adj}} \Rightarrow \tan(60°) = \frac{\sqrt{3}}{1} \text{ or } \sqrt{3}$$

$$\sin(B) = \frac{\text{opp}}{\text{hyp}} \Rightarrow \sin(30°) = \frac{1}{2}$$

$$\cos(B) = \frac{\text{adj}}{\text{hyp}} \Rightarrow \cos(30°) = \frac{\sqrt{3}}{2}$$

$$\tan(B) = \frac{\text{opp}}{\text{adj}} \Rightarrow \tan(30°) = \frac{1}{\sqrt{3}} \text{ or } \frac{\sqrt{3}}{3}$$

- Consider a right-angled isosceles ΔEFG with equal sides of 1 unit.



$$(EG)^2 = (EF)^2 + (FG)\,2 \quad \text{(using Pythagoras' theorem)}$$
$$(EG)^2 = 1^2 + 1^2$$
$$EG = \sqrt{2}$$

- Using ΔEFG, the following exact values are obtained:

$$\sin(E) = \frac{\text{opp}}{\text{hyp}} \Rightarrow \sin(45°) = \frac{1}{\sqrt{2}} \text{ or } \frac{\sqrt{2}}{2}$$
$$\cos(E) = \frac{\text{adj}}{\text{hyp}} \Rightarrow \cos(45°) = \frac{1}{\sqrt{2}} \text{ or } \frac{\sqrt{2}}{2}$$
$$\tan(E) = \frac{\text{opp}}{\text{adj}} \Rightarrow \tan(45°) = \frac{1}{1} \text{ or } 1$$

Summary of exact values

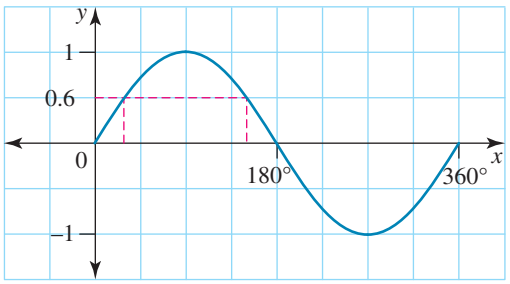
θ	30°	45°	60°
sin(θ)	$\frac{1}{2}$	$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$
cos(θ)	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$	$\frac{1}{2}$
tan(θ)	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$	1	$\sqrt{3}$

eles-5012

15.7.2 Solving trigonometric equations

Solving trigonometric equations graphically

- Because of the periodic nature of circular functions, there are infinitely many solutions to unrestricted trigonometric equations.
- Equations are usually solved within a particular domain (x-values), to restrict the number of solutions.
- The sine graph shown displays the solutions between 0° and 360° for the equation $\sin(x) = 0.6$.



In the graph, it can clearly be seen that there are two solutions to this equation, which are approximately $x = 37°$ and $x = 143°$.

- It is difficult to obtain accurate answers from a graph. More accurate answers can be obtained using technology.

Solving trigonometric equations algebraically

- Exact values can be found for some trigonometric equations using the table in section 15.7.1.

WORKED EXAMPLE 14 Solving trigonometric equations using exact values

Solve the following equations.

a. $\sin(x) = \frac{\sqrt{3}}{2}, x \in [0°, 360°]$

b. $\cos(2x) = -\frac{1}{\sqrt{2}}, x \in [0°, 360°]$

THINK

a. 1. The inverse operation of sine is $\sin^{-1}$.

2. The first solution in the given domain from the table in subsection 15.7.1 is $x = 60°$. Since sine is positive in the first and second quadrants, another solution must be $x = 180° - 60° = 120°$.

b. 1. The inverse operation of cosine is $\cos^{-1}$.

2. From the table of values, $\cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45°$.
Cosine is negative in the second and third quadrants, which gives the first two solutions to the equation as $180° - 45°$ and $180° + 45°$.

3. Solve for x by dividing by 2.

4. Since the domain in this case is [0°, 360°] and the period has been halved, there must be four solutions altogether.
The other two solutions can be found by adding the period onto each solution.

WRITE

a. $x = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

There are two solutions in the given domain, $x = 60°$ and $x = 120°$.

b. $2x = \cos^{-1}\left(\frac{-1}{\sqrt{2}}\right)$

$2x = 135°, 225°$

$x = 67.5°, 112.5°$

$\text{Period} = \frac{360°}{2} = 180°$

$x = 67.5° + 180°, 112.5° + 180°$

$x = 67.5°, 112.5°, 247.5°, 292.5°$

TI | THINK

a.

In a new problem, on a Calculator page, complete the entry line as:

$$\text{solve}\left(\sin(x)=\frac{\sqrt{3}}{2}, x\right) \mid$$

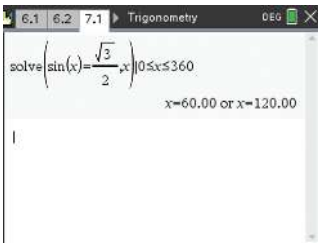
$0 \le x \le 360$

Then press ENTER.

Note that the calculator is set to the degrees mode.

DISPLAY/WRITE

a.



$\sin(x)=\frac{\sqrt{3}}{2}$ for $x \in [0°, 360°]$

$\Rightarrow x = 60°$ or $120°$

b.

On a Calculator page, complete the entry line as:

$$\text{solve}\left(\cos(2x)=-\frac{1}{\sqrt{2}}, x\right) \mid$$

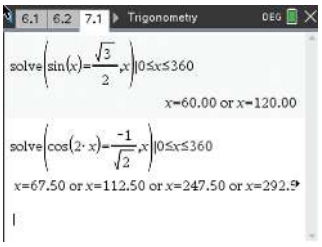
$0 \le x \le 360$

Then press ENTER.

Note that the calculator is set to the degrees mode.

DISPLAY/WRITE

b.



$\cos(2x)=-\frac{1}{\sqrt{2}}$ for $x \in [0°, 360°]$

$\Rightarrow x = 67.5°, 112.5°, 247.5°$ or $292.5°$

CASIO | THINK

a.

On the Main screen, complete the entry line as:

$$\text{solve}\left(\sin(x)=\frac{\sqrt{3}}{2}, x\right) \mid$$

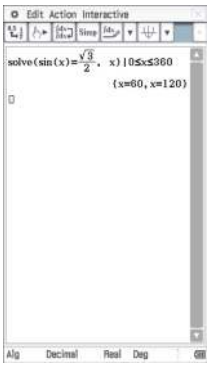
$0 \le x \le 360$

Then press EXE.

Note that the calculator is set to the degrees mode.

DISPLAY/WRITE

a.



$\sin(x)=\frac{\sqrt{3}}{2}$ for $x \in [0°, 360°]$

$\Rightarrow x = 60°$ or $120°$

b.

On the Main screen, complete the entry line as:

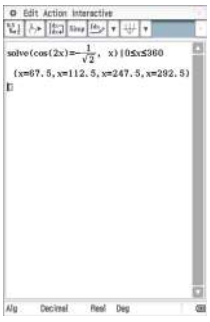
$$\text{solve}\left(\cos(2x)=-\frac{1}{\sqrt{2}}, x\right) \mid$$

$0 \le x \le 360$

Then press EXE.

Note that the calculator is set to the degrees mode.

b.



$\cos(2x)=-\frac{1}{\sqrt{2}}$ for $x \in [0°, 360°]$

$\Rightarrow x = 67.5°, 112.5°, 247.5°$ or $292.5°$

DISCUSSION

Explain why sine and cosine functions can be used to model situations that occur in nature such as tide heights and sound waves.

on Resources

eWorkbook Topic 15 Workbook (worksheets, code puzzle and project) (ewbk-13308)

Interactivities Exact values of trigonometric functions (int-4816)
Solving trigonometric equations graphically (int-4822)

Exercise 15.7 Solving trigonometric equations

learnon

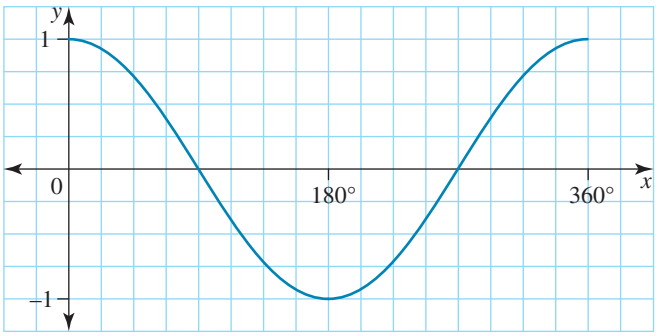
15.7 Quick quiz on | 15.7 Exercise

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 3, 6, 9, 13, 16	2, 4, 7, 10, 14, 17	5, 8, 11, 12, 15, 18

Fluency

For questions 1 and 2, use the graph to determine approximate answers to the equations for the domain $0 \leq x \leq 360°$. Check your answers using a calculator.



1. a. $\cos(x) = 0.9$ b. $\cos(x) = 0.3$

2. a. $\cos(x) = -0.2$ b. $\cos(x) = -0.6$

For questions 3 to 8, solve the equations for the domain $0° \leq x \leq 360°$.

3. a. $\sin(x) = \frac{1}{2}$ b. $\sin(x) = \frac{\sqrt{3}}{2}$

4. a. $\cos(x) = -\frac{1}{2}$ b. $\cos(x) = -\frac{1}{\sqrt{2}}$

5. a. $\sin(x) = 1$ b. $\cos(x) = -1$

6. a. $\sin(x) = -\frac{1}{2}$ b. $\sin(x) = -\frac{1}{\sqrt{2}}$

7. a. $\cos(x) = \frac{\sqrt{3}}{2}$ b. $\cos(x) = -\frac{\sqrt{3}}{2}$

8. a. $\sin(x) = 1$ b. $\cos(x) = 0$

Understanding

WE14 For questions **9** to **12**, solve the equations for the given values of x.

9. **a.** $\sin(2x) = \frac{\sqrt{3}}{2}, x \in [0°, 360°]$ **b.** $\cos(2x) = -\frac{\sqrt{3}}{2}, x \in [0°, 360°]$

10. **a.** $\tan(2x) = \frac{1}{\sqrt{3}}, x \in [0°, 360°]$ **b.** $\sin(3x) = -\frac{1}{2}, x \in [0°, 180°]$

11. **a.** $\sin(4x) = -\frac{1}{2}, x \in [0°, 180°]$ **b.** $\sin(3x) = -\frac{1}{\sqrt{2}}, x \in [-180°, 180°]$

12. **a.** $\tan(3x) = -1, x \in [0°, 90°]$ **b.** $\cos(3x) = 0, x \in [0°, 360°]$

Reasoning

13. Solve the following equations for $x \in [0°, 360°]$.

a. $2\sin(x) - 1 = 0$ **b.** $2\cos(x) = \sqrt{3}$

14. Solve the following equations for $x \in [0°, 360°]$.

a. $\sqrt{2}\cos(x) - 1 = 0$ **b.** $\tan(x) + 1 = 0$

15. Sam measured the depth of water at the end of the Intergate jetty at various times on Thursday 13 August 2020. The table below provides her results.

Time	6 am	7	8	9	10	11	12 pm	1	2	3	4	5	6	7	8	9
Depth (m)	1.5	1.8	2.3	2.6	2.5	2.2	1.8	1.2	0.8	0.5	0.6	1.0	1.3	1.8	2.2	2.5

a. Plot the data.

b. Determine:

i. the period

ii. the amplitude.

c. Sam fishes from the jetty when the depth is a maximum. Specify these times for the next 3 days.

d. Sam's mother can moor her yacht when the depth is above 1.5 m. Determine during what periods she can moor the yacht on Sunday 16 January.



Problem solving

16. Solve:

a. $\sqrt{3}\sin(x°) = \cos(x°)$ for $0° \leq x \leq 360°$

b. $2\sin(x°) + \cos(x°) = 0$ for $0° \leq x \leq 360°$.

17. Solve $2\sin^2(x°) + 3\sin(x°) - 2 = 0$ for $0° \leq x \leq 360°$.

18. The grad (^{g}) is another measurement used when measuring the size of angles. A grad is equivalent to $\frac{1}{400}$ of a full circle. Write each of the following as grads (1 grad is written as 1^g).

a. 90° **b.** 180° **c.** 270° **d.** 360°

LESSON
15.8 Review

15.8.1 Topic summary

Sine rule
- Connects two sides with the two opposite angles in a triangle.
- $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
- Used to solve triangles:
  - Two angles and one side.
  - Two sides and an angle opposite one of these sides.

Area of a triangle
- Given two sides and the included angle:
  $\text{Area} = \frac{1}{2}ab\sin(C)$
- If the lengths of all the sides of the triangle are known but none of the angles, apply the cosine rule to calculate one of the angles.
- Substitute the values of the angle and the two adjacent sides into the area of a triangle formula.

Cosine rule
- Connects three sides and one angle of a triangle.
  $a^2 = b^2 + c^2 - 2bc\cos(A)$
  Used to solve triangles given:
  - three sides, or
  - two sides and the included angle.

Ambiguous case
- When using the sine rule to calculate an angle, there may be two answers.
- The ambiguous case may occur when determining the angle opposite the larger side.
- Always c heck that the three angles add to 180°.

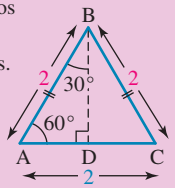
e.g. In the triangle ABC, $a = 10$, $b = 6$ and $B = 30°$, using the sine rule, $A = 56°$ or $(180 - 56)°$.
Angles in triangle would be:
$A = 56°$, $B = 30°$ giving $C = 94°$
$A = 124°$, $B = 30°$ giving $C = 26°$
Two triangles are possible, so the ambiguous case exists.

TRIGONOMETRY II (10A)

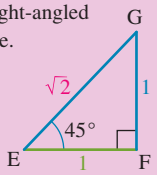
Trigonometric equations
- Infinite number of solutions, so domain is normally restricted:
  e.g. $0° \le x \le 180°$
- Equations can be solved:
  - graphically — not very accurate
  - using technology
  - algebraically, using the exact values.

e.g. $\sin\alpha = -\frac{1}{2}$, $0° \le x \le 360°$
from exact values: $\sin 30° = \frac{1}{2}$
sine is negative in 3rd and 4th quadrants and the angle is from the x-axis
$\alpha = (180 + 30)°$ or $(360 - 30)°$
$\alpha = 210°$ or $330°$

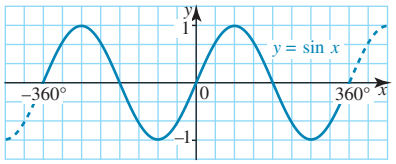
Exact values
- Exact trig ratios can be found using triangles.
- For 30° and 60° use the equilateral triangle.
- For 45°, use a right-angled isosceles triangle.



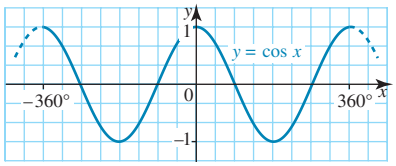
e.g. $\sin 30° = \frac{1}{2}$
$\tan 45° = 1$
$\cos 60° = \frac{1}{2}$



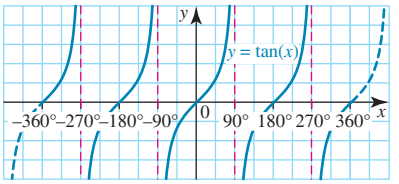
Trigonometric graphs
- Trigonometric graphs repeat themselves continuously in cycles.
- Period: horizontal distance between repeating peaks or troughs.
- Amplitude: half the distance between the maximum and minimum values.



$y = \sin x$, Period = 360°, Amplitude = 1

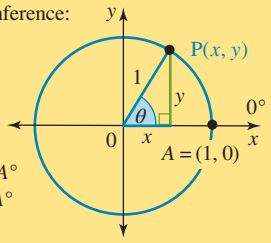


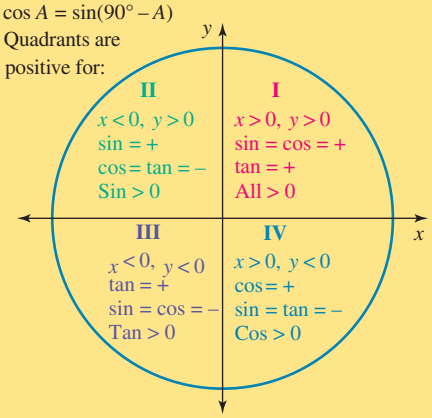
$y = \cos x$, Period = 360°, Amplitude = 1



$y = \tan x$, Period = 180°, Amplitude = undefined.

Unit circle
- Equation of the unit circle: $x^2 + y^2 = 1$
- Radius of length 1 unit.
- For any point on circumference:
  $x = \cos\theta$
  $y = \sin\theta$
- $\tan\theta = \frac{y}{x} = \frac{\sin\theta}{\cos\theta}$
- $\sin(180 - A)° = \sin A°$
  $\cos(180 - A)° = -\cos A°$
  $\tan(180 - A)° = -\tan A°$
- $\sin A = \cos(90° - A)$
  $\cos A = \sin(90° - A)$
- Quadrants are positive for:





15.8.2 Success criteria

Tick the column to indicate that you have completed the lesson and how well you have understood it using the traffic light system.

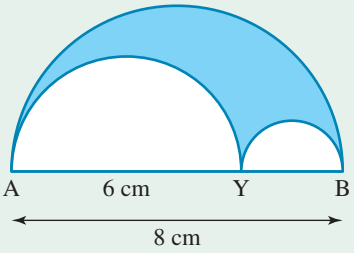
(**Green:** I understand; **Yellow:** I can do it with help; **Red:** I do not understand)

Lesson	Success criteria			
15.2	I can apply the sine rule to evaluate angles and sides of triangles.			
	I can recognise when the ambiguous case of the sine rule exists.			
15.3	I can apply the cosine rule to calculate a side of a triangle.			
	I can apply the cosine rule to calculate the angles of a triangle.			
15.4	I can calculate the area of a triangle, given two sides and the included angle.			
	I can calculate the area of a triangle, given the three sides.			
15.5	I can determine in which quadrant an angle lies.			
	I can use and interpret the relationship between a point on the unit circle and the angle made with the positive x-axis.			
	I can use the unit circle to determine approximate trigonometric ratios for angles greater than 90°.			
	I can use and interpret the relationships between supplementary and complementary angles			
15.6	I can sketch the graphs of the sine, cosine and tangent graphs.			
	I can determine the amplitude of a given trigonometric function.			
	I can determine the period of a given trigonometric function.			
15.7	I can apply the exact values of sin, cos and tan for 30°, 45° and 60° angles.			
	I can solve trigonometric equations graphically for a given domain.			
	I can solve trigonometric equations algebraically, using exact values, for a given domain.			

15.8.3 Project

What's an arbelos?

As an introduction to this task, you are required to complete the following construction. The questions that follow require the application of measurement formulas and an understanding of semicircles related to this construction.



1. **Constructing an arbelos**
   - Rule a horizontal line AB 8 cm long.
   - Determine the midpoint of the line and construct a semicircle on top of the line with AB as the diameter.
   - Mark Y as a point on AB such that AY = 6 cm.
   - Determine the midpoint of AY and draw a small semicircle inside the larger semicircle with AY as the diameter.
   - Determine the midpoint of YB and construct a semicircle (also inside the larger semicircle) with a diameter YB.

The shape enclosed by the three semicircles is known as an arbelos. The word, in Greek, means 'shoemaker's knife' as it resembles the blade of a knife used by cobblers. The point Y is not fixed and can be located anywhere along the diameter of the larger semicircle, which can also vary in size.



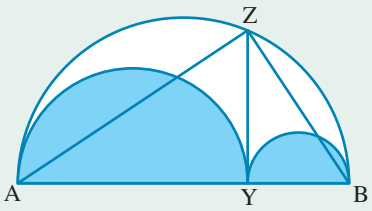
2. **Perimeter of an arbelos**
   The perimeter of an arbelos is the sum of the arc length of the three semicircles. Perform the following calculations, leaving each answer in terms of π.
   a. Calculate the arc length of the semicircle with diameter AB.
   b. Calculate the arc length of the semicircle with diameter AY.
   c. Calculate the arc length of the semicircle on diameter YB.
   d. Compare the largest arc length with the two smaller arc lengths. What do you conclude?
3. We can generalise the arc length of an arbelos. The point Y can be located anywhere on the line AB, which can also vary in length.
   Let the diameter AB be d cm, AY be d_1 cm and YB be d_2 cm.
   Prove that your conclusion from question **2d** holds true for any value of d, where $d_1 + d_2 = d$.

4. Area of an arbelos

The area of an arbelos may be treated as the area of a composite shape.

a. Using your original measurements, calculate the area of the arbelos you drew in question **1**. Leave your answer in terms of π.

The area of the arbelos can also be calculated using another method. We can draw the common tangent to the two smaller semicircles at their point of contact and extend this tangent to the larger semicircle. It is said that the area of the arbelos is the same as the area of the circle constructed on this common tangent as diameter.



YZ is the common tangent.

Triangles AYZ, BYZ and AZB are all right-angled triangles. We can use Pythagoras' theorem, together with a set of simultaneous equations, to determine the length of the tangent YZ.

b. Complete the following.

In ΔAYZ, $AZ^2 = AY^2 + YZ^2$

$= 6^2 + YZ^2$

In ΔBYZ, $BZ^2 = BY^2 + YZ^2$

$= \ldots\ldots\ldots\ldots + YZ^2$

Adding these two equations,

$AZ^2 + BZ^2 = \ldots\ldots\ldots\ldots + \ldots\ldots\ldots\ldots$

$AZ^2 + BZ^2 = AB^2$

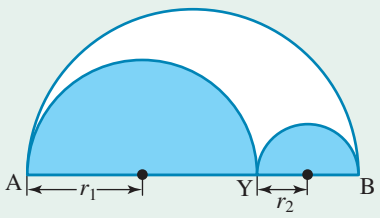
But, in ΔAZB $= \ldots\ldots\ldots\ldots$

$\ldots\ldots\ldots\ldots + \ldots\ldots\ldots\ldots$

So, $YZ = \ldots\ldots\ldots\ldots$ (Leave your answer in surd form.)

c. Now calculate the area of the circle with diameter YZ.
Is your answer the same as that calculated in part **a**?

The area of an arbelos can be generalised.
Let the radii of the two smaller semicircles be r_1 and r_2.



5. Develop a formula for the area of the arbelos in terms of r_1 and r_2.
Demonstrate the use of your formula by checking your answer to question **4a**.

Resources

eWorkbook Topic 15 Workbook (worksheets, code puzzle and project) (ewbk-13308)

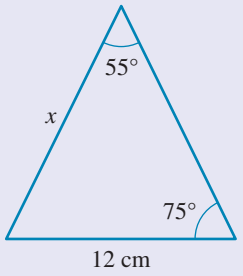
Interactivities Crossword (int-2884)
Sudoku puzzle (int-3895)

Exercise 15.8 Review questions

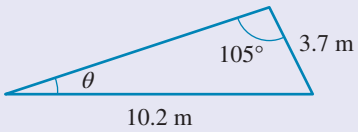
learnon

Fluency

1. Calculate the value of x, correct to 2 decimal places.

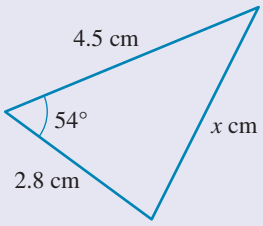


2. Calculate the value of θ, correct to the nearest minute.

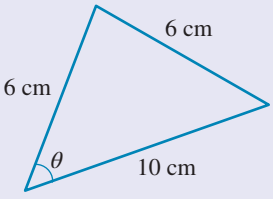


3. Determine all unknown sides (correct to 2 decimal places) and angles (correct to the nearest minute) of triangle ABC, given $a = 25$ m, $A = 120°$ and $B = 50°$.

4. Calculate the value of x, correct to 2 decimal places.



5. Calculate the value of θ, correct to the nearest degree.



6. A triangle has sides of length 12 m, 15 m and 20 m. Calculate the magnitude (size) of the largest angle, correct to the nearest minute.

7. A triangle has two sides of 18 cm and 25 cm. The angle between the two sides is 45°.
   Calculate, correct to 2 decimal places:
   a. its area
   b. the length of its third side.

8. If an angle of $\theta = 290°$ was represented on the unit circle, state which quadrant the triangle to show this would be drawn in.

9. On the unit circle, draw an appropriate triangle for the angle 110° in quadrant 2.
   a. Determine the value of sin(110°) and cos(110°), correct to 2 decimal places.
   b. Determine the value of tan(110°), correct to 2 decimal places.

10. MC The value of sin(53°) is equal to:
   A. cos(53°) B. cos(37°) C. sin(37°) D. tan(53°)

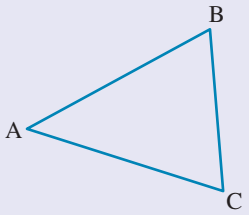
11. Simplify $\dfrac{\sin(53°)}{\sin(37°)}$.

12. Draw a sketch of $y = \sin(x)$ from $0° \leq x \leq 360°$.

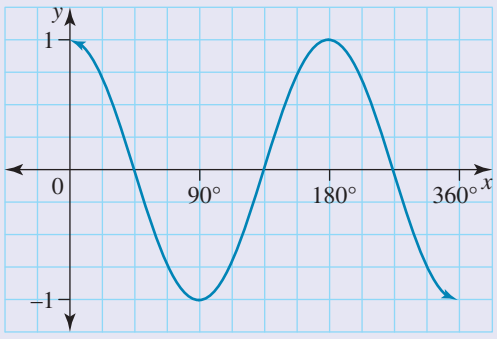
13. Draw a sketch of $y = \cos(x)$ from $0° \leq x \leq 360°$.

14. Draw a sketch of $y = \tan(x)$ from $0° \leq x \leq 360°$.

15. Label the triangle shown so that $\dfrac{x}{\sin(46°)} = \dfrac{y}{\sin(68°)}$.



16. State the period and amplitude of each of the following graphs.
   a. $y = 2\sin(3x)$
   b. $y = -3\cos(2x)$
   c.



17. Sketch the following graphs.

a. $y = 2\sin(x), x \in [0°, 360°]$ **b.** $y = \cos(2x), x \in [-180°, 180°]$

18. Use technology to write down the solutions to the following equations for the domain $0° \leq x \leq 360°$ to 2 decimal places.

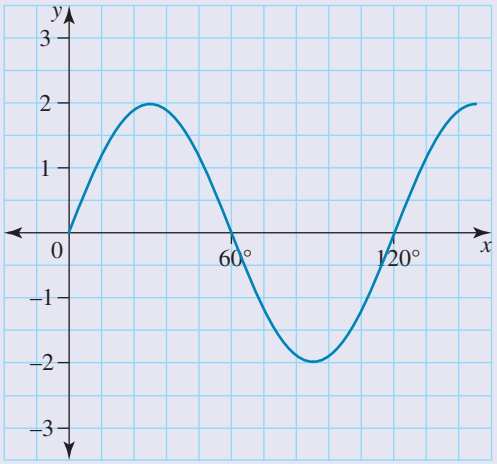
a. $\sin(x) = -0.2$ **b.** $\cos(2x) = 0.7$ **c.** $3\cos(x) = 0.1$ **d.** $2\tan(2x) = 0.5$

19. Solve each of the following equations.

a. $\sin(x) = \frac{1}{2}, x \in [0°, 360°]$ **b.** $\cos(x) = \frac{\sqrt{3}}{2}, x \in [0°, 360°]$

c. $\cos(x) = \frac{1}{\sqrt{2}}, x \in [0°, 360°]$ **d.** $\sin(x) = \frac{1}{\sqrt{2}}, x \in [0°, 360°]$

20. MC The equation that represents the graph shown could be:



A. $y = 3\sin(2x)$ **B.** $y = 2\cos(3x)$ **C.** $y = 3\cos(2x)$ **D.** $y = 2\sin(3x)$

21. **a.** Use technology to help sketch the graph of $y = 2\sin(2x) - 3$.

b. Write down the period and the amplitude of the graph in part **a**.

Problem solving

22. Sketch the graphs of each of the following, stating:

i. the period

ii. the amplitude.

a. $y = 2\cos(2x), x \in [0°, 360°]$

b. $y = 3\sin(4x), x \in [0°, 180°]$

c. $y = -2\cos(3x), x \in [-60°, 60°]$

d. $y = 4\sin(2x), x \in [-90°, 90°]$

23. Solve each of the following equations for the given values of x.

a. $\cos(2x) = \frac{\sqrt{3}}{2}, x \in [0°, 360°]$ **b.** $\sin(3x) = \frac{1}{2}, x \in [-90°, 90°]$

c. $\sin(2x) = \frac{1}{\sqrt{2}}, x \in [0°, 360°]$

d. $\cos(3x) = -\frac{1}{\sqrt{2}}, x \in [0°, 360°]$

e. $\sin(4x) = 0, x \in [0°, 180°]$

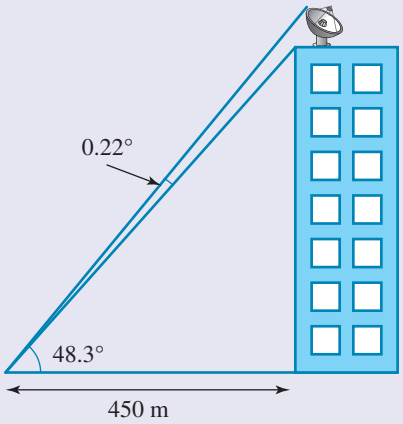
f. $\tan(4x) = -1, x \in [0°, 180°]$

24. Solve the following for $x \in [0°, 360°]$.

a. $2\cos(x) - 1 = 0$ **b.** $2\sin(x) = -\sqrt{3}$ **c.** $-\sqrt{2}\cos(x) + 1 = 0$ **d.** $\sqrt{2}\sin(x) + 1 = 0$

25. Sketch the graph of $y = \tan(2x), x \in [0°, 180°]$. Write down the period, amplitude and the equations of any asymptotes.

26. A satellite dish is placed on top of an apartment building as shown in the diagram. Determine the height of the satellite dish in metres, correct to 2 decimal places.



27. Australian power points supply voltage, V, in volts, where $V = 240\,(\sin 18\,000t)$ and t is measured in seconds.

t	V
0.000	
0.005	
0.010	
0.015	
0.020	
0.025	
0.030	
0.035	
0.040	

a. Copy and complete the table and sketch the graph, showing the fluctuations in voltage over time.

b. State the times at which the maximum voltage output occurs.

c. Determine how many seconds there are between times of maximum voltage output.

d. Determine how many periods (or cycles) there are per second.



To test your understanding and knowledge of this topic, go to your learnON title at www.jacplus.com.au and complete the **post-test**.

Answers

Topic 15 Trigonometry II (10A)

15.1 Pre-test

1. B
2. 9.06 m
3. 9.35 cm
4. A
5. 19.67
6. 7.2 m
7. B
8. 3rd quadrant
9. 0.5
10. D
11. D
12. B
13. A
14. C
15. 270°

15.2 The sine rule

1. 44°58′, 77°2′, 13.79
2. 39°18′, 38°55′, 17.21
3. 70°, 9.85, 9.40
4. 33°, 38.98, 21.98
5. 19.12
6. $C = 51°, b = 54.66, c = 44.66$
7. $A = 60°, b = 117.11, c = 31.38$
8. $B = 48°26', C = 103°34', c = 66.26$; or $B = 131°34', C = 20°26', c = 23.8$
9. 24.17
10. B, C
11. $A = 73°15', b = 8.73$; or $A = 106°45', b = 4.12$
12. 51.90 or 44.86
13. $C = 110°, a = 3.09, b = 4.64$
14. $B = 38°, a = 3.36, c = 2.28$
15. $B = 33°33', C = 121°27', c = 26.24$; or $B = 146°27', C = 8°33', c = 4.57$
16. 43.62 m
17. $h = 7.5$ cm
18. 113 km
19. 8.68 m
20. Yes, she needs 43 m altogether.
21. a. 6.97 m b. 4 m
22. a. 13.11 km b. N20°47′W
23. a. 8.63 km b. 6.5 km/h c. 9.90 km
24. 22.09 km from A and 27.46 km from B.
25. a. 7.3 km b. 282°3′

15.3 The cosine rule

1. 7.95
2. 55.22
3. 23.08, 41°53′, 23°7′
4. 28°57′
5. 88°15′
6. $A = 61°15', B = 40°, C = 78°45'$
7. 2218 m
8. a. 12.57 km b. S35°1′E
9. a. 35°6′ b. 6.73 m^2
10. 23°
11. 89.12 m
12. a. 130 km b. S22°12′E
13. 28.5 km
14. 74.3 km
15. 70°49′
16. a. 8.89 m b. 77°0′ c. $x = 10.07$ m
17. 1.14 km/h
18. ∠CAB = 34.65°, ∠ABC = 84.83° and ∠BCA = 60.52°
19. a. 4.6637 m b. 55.93°

15.4 Area of triangles

1. 12.98
2. 38.14
3. 212.88
4. $A = 32°4', B = 99°56'$, area = 68.95 cm^2
5. $A = 39°50', B = 84°10'$, area = 186.03 m^2
6. $A = 125°14', B = 16°46'$, area = 196.03 mm^2
7. C
8. 14.98 cm^2
9. 570.03 mm^2
10. 2.15 cm^2
11. B
12. 3131.41 mm^2
13. 610 cm^2
14. a. 187.5 cm^2 b. 15.03 cm^2
15. 17 goldfish
16. C
17. B
18. a. Area = 69.63 cm^2
    b. The dimensions are 12.08 cm and 6.96 cm.
19. 17 kg
20. 52.2 hectares
21. 175 m^3
22. 22.02 m^2
23. a. 29.44 km^2 b. 8.26 km c. 41°
    d. 24° e. 3.72 km f. 10.11 km^2
    g. 39.55 km^2
24. 374.12 cm^2

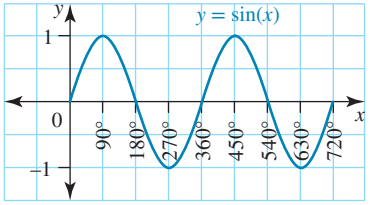
15.5 The unit circle

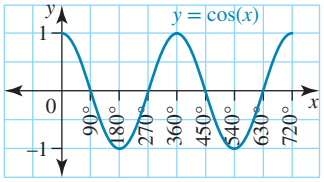
1. **a.** 1st **b.** 2nd **c.** 4th **d.** 3rd **e.** 2nd **f.** 3rd
2. A
3. D
4. **a.** 0.34 **b.** 0.94 **c.** −0.17 **d.** 0.98
5. **a.** −0.64 **b.** 0.77 **c.** −0.57 **d.** −0.82
6. **a.** 1 **b.** 0 **c.** 0 **d.** −1
7. **a.** −1 **b.** 0 **c.** 0 **d.** 1
8. **a.** 0.87 **b.** 0.50
9. **a.** 30°
   b. −0.87
   c. $\cos(150°) = -\cos(30°)$
   d. 0.5
   e. $\sin(150°) = \sin(30°)$
10. **a.** 30°
    b. −0.87
    c. $\cos(210°) = -\cos(30°)$
    d. −0.50
    e. $\sin(210°) = -\sin(30°)$
11. **a.** 30°
    b. 0.87
    c. $\cos(330°) = -\cos(30°)$
    d. −0.50
    e. $\sin(330°) = -\sin(30°)$
12. **a.** 0.34 **b.** 0.94
    c. 0.36 **d.** 0.36
    e. They are equal.
13. **a.** 0.71 **b.** −0.71
    c. −1 **d.** −1
    e. They are equal. **f.** $\tan(135°) = -\tan(45°)$
14. **a.** −0.64 **b.** −0.77
    c. 0.84 **d.** 0.83
    e. They are approximately equal.
    f. $\tan(220°) = \tan(40°)$
15. **a.** −0.87 **b.** 0.5
    c. −1.73 **d.** −1.74
    e. They are approximately equal.
    f. $\tan(300°) = -\tan(60°)$
16. D

17, 18, 19. Sample responses can be found in the worked solutions in the online resources.

20. **a.** $\sqrt{1-p^2}$ **b.** p
    c. $-\sqrt{1-p^2}$ **d.** p
21. 0
22. 0

15.6 Trigonometric functions

1. See the table at the bottom of the page.*
2. 

3. 360°
4. **a.** 0.7 **b.** 0.8 **c.** 0.3 **d.** −0.3
5. **a.** 0 **b.** 0.9 **c.** −0.2 **d.** −0.9
6. **a.** 64°, 116°, 424°, 476°
   b. 244°, 296°, 604°, 656°
   c. 44°, 136°, 404°, 496°
7. **a.** 210°, 330°, 570°, 690°
   b. 233°, 307°, 593°, 667°
   c. 24°, 156°, 384°, 516°
8. See the table at the bottom of the page.*
9. 

10. The graph would continue with the cycle.
11. It is a very similar graph with the same shape; however, the sine graph starts at (0, 0), whereas the cosine graph starts at (0, 1).
12. **a.** 0.7 **b.** −0.9 **c.** −1 **d.** 0.9
13. **a.** −0.5 **b.** −0.8 **c.** 0.8 **d.** −0.9
14. **a.** 120°, 240°, 480°, 600°
    b. 37°, 323°, 397°, 683°
    c. 46°, 314°, 406°, 674°
15. **a.** 127°, 233°, 487°, 593°
    b. 26°, 334°, 386°, 694°
    c. 154°, 206°, 514°, 566°

*1.

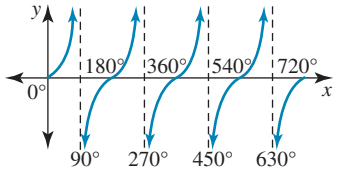
x	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\sin(x)$	0	0.5	0.87	1	0.87	0.5	0	−0.5°	−0.87	−1	−0.87	−0.5	0
x	390°	420°	450°	480°	510°	540°	570°	600°	630°	660°	690°	720°	
$\sin(x)$	0.5	0.87	1	0.87	0.5	0	−0.5	−0.87	−1	−0.87	−0.5	0	

*8.

x	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\cos(x)$	1	0.87	0.5	0	−0.5	−0.87	−1	−0.87	−0.5	0	0.5	0.87	1
x	390°	420°	450°	480°	510°	540°	570°	600°	630°	660°	690°	720°	
$\cos(x)$	0.87	0.5	0	−0.5	−0.87	−1	−0.87	−0.5	0	0.5	0.87	1	

16. See the table at the bottom of the page.*

17. $y = \tan(x)$



18. The graph would continue repeating every 180° as above.

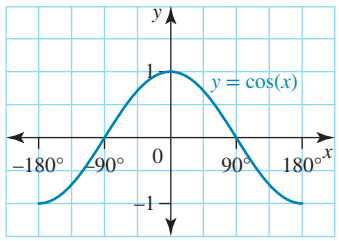
19. Quite different. $y = \tan(x)$ has undefined values (asymptotes) and repeats every 180° rather than 360°. It also gives all y-values, rather than just values between -1 and 1.

20. **a.** 1.7 **b.** -1 **c.** -1.2 **d.** 0.8

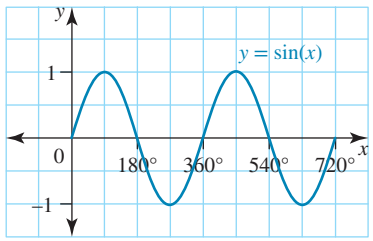
21. **a.** -0.8 **b.** 1.2 **c.** -0.2 **d.** 1

22. **a.** 45°, 225°, 405°, 585°
b. 56°, 236°, 416°, 596°
c. 158°, 338°, 518°, 698°
d. 117°, 297°, 477°, 657°
e. 11°, 191°, 371°, 551°
f. 135°, 315°, 495°, 675°

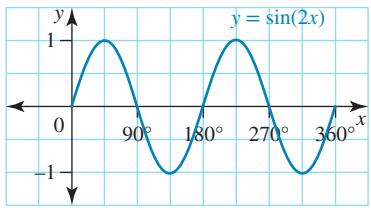
23. **a.** **i.** 360°
ii. 1
iii.



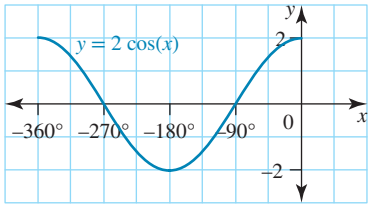
b. **i.** 360°
ii. 1
iii.



24. **a.** **i.** 180°
ii. 1
iii.



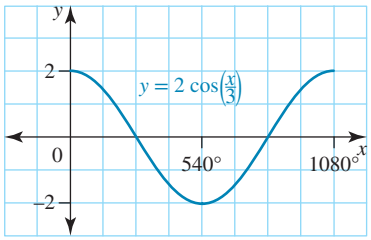
b. **i.** 360°
ii. 2
iii.



25. **a.** **i.** 180° **ii.** 3
b. **i.** 120° **ii.** 4
c. **i.** 720° **ii.** 2
d. **i.** 1440° **ii.** $\frac{1}{2}$
e. **i.** 360° **ii.** 1
f. **i.** 180° **ii.** 1

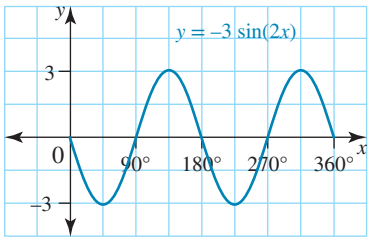
26. **a.** C **b.** A **c.** D

27. **a.**



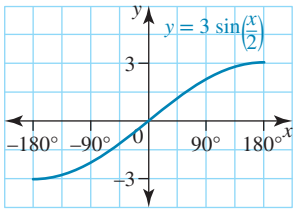
Period = 1080°
Amplitude = 2

b.



Period = 180°
Amplitude = 3

c.

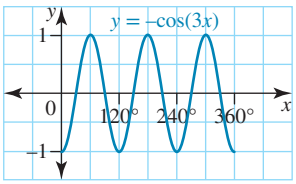


Period = 720°
Amplitude = 3

*16.

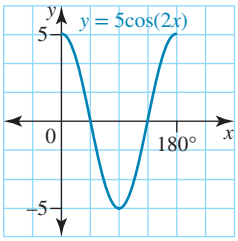
x	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\tan(x)$	0	0.58	1.73	undef.	−1.73	−0.58	0	0.58	1.73	undef.	−1.73	−0.58	0
x	390°	420°	450°	480°	510°	540°	570°	600°	630°	660°	690°	720°	
$\tan(x)$	0.58	1.73	undef.	−1.73	−0.58	0	0.58	1.73	undef.	−1.73	−0.58	0	

d.



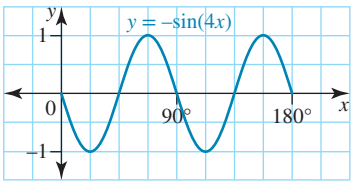
Period = 120°
Amplitude = 1

e.



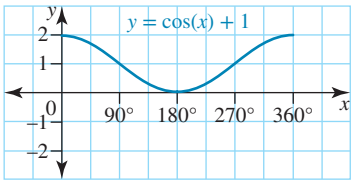
Period = 180°
Amplitude = 5

f.

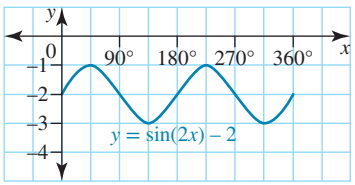


Period = 90°
Amplitude = 1

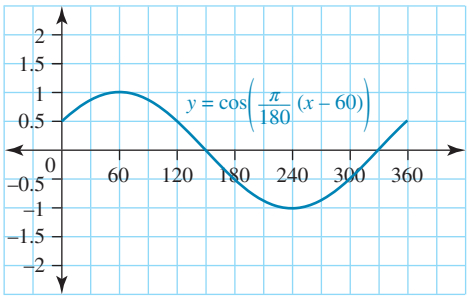
28. a.



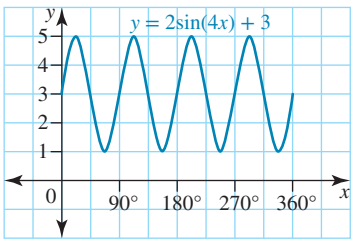
b.



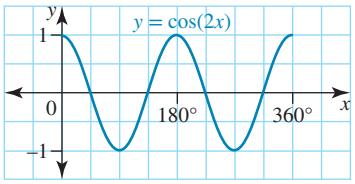
c.



d.



29. a.



i. –1 **ii.** 1

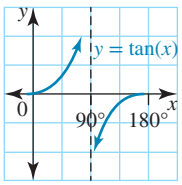
b. Max value = 3, min value = 1

c. Max value of $\sin(x) = 1$; hence, max value of $y = 2 \times 1 + 3 = 5$
Min value of $\sin(x) = -1$; hence, min value of $y = 2 \times -1 + 3 = 1$

30. a.

x	0	30°	60°	90°	120°	150°	180°
y	0	$\frac{\sqrt{3}}{3}$	$\sqrt{3}$	undef.	$-\sqrt{3}$	$-\frac{\sqrt{3}}{3}$	0

b.

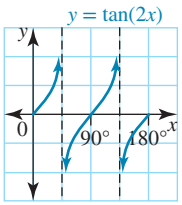


c. At $x = 90°$, y is undefined.

d. $x = 270°$

e. The period = 180° and the amplitude is undefined.

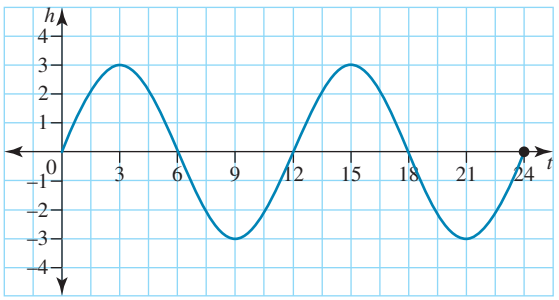
31. a.



b. $x = 45°$ and $x = 135°$

c. The period = 90° and the amplitude is undefined.

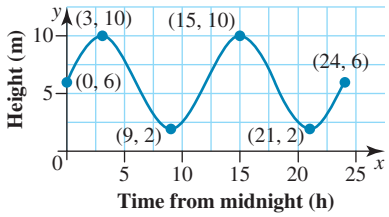
32. a.



b. 3 metres

c. –2.6 metres

33. a.



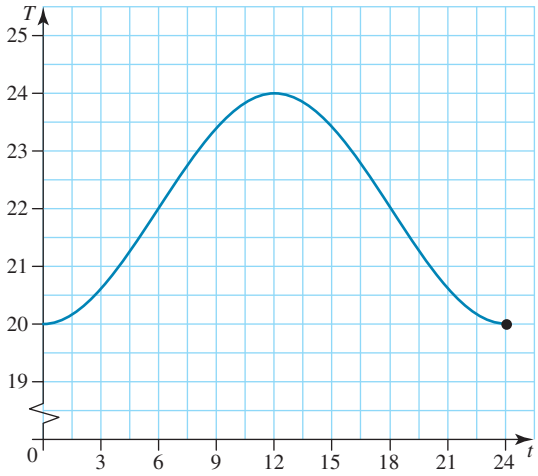
b. 10 metres

c. 2 metres

d. 2.54 metres

34. a. 22 °C

b.



c. Coolest 20 °C, warmest 24 °C

15.7 Solving trigonometric equations

1. Calculator answers

a. 25.84°, 334.16° **b.** 72.54°, 287.46°

2. a. 101.54°, 258.46° **b.** 126.87°, 233.13°

3. a. 30°, 150° **b.** 60°, 120°

4. a. 120°, 240° **b.** 135°, 225°

5. a. 90° **b.** 180°

6. a. 210°, 330° **b.** 225°, 315°

7. a. 30°, 330° **b.** 150°, 210°

8. a. 90° **b.** 90°, 270°

9. a. 30°, 60°, 210°, 240° **b.** 75°, 105°, 255°, 285°

10. a. 15°, 105°, 195°, 285°

b. 70°, 110°

11. a. 52.5°, 82.5°, 142.5°, 172.5°

b. −165°, −135°, −45°, −15°, 75°, 105°

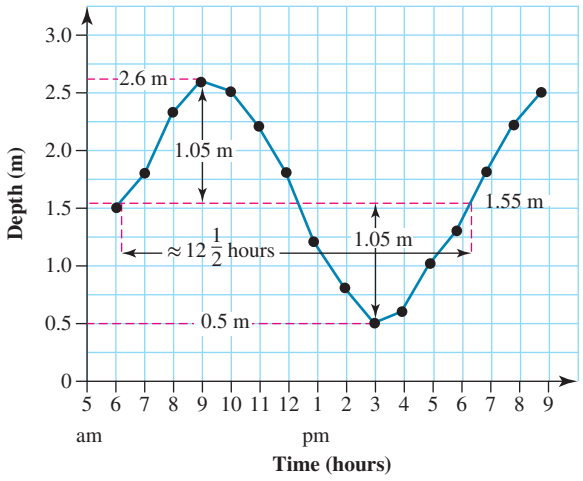
12. a. 45°

b. 30°, 90°, 150°, 210°, 270°, 330°

13. a. 30°, 150° **b.** 30°, 330°

14. a. 45°, 315° **b.** 135°, 315°

15. a.



b. i. $12\frac{1}{2}$ hours **ii.** 1.05 m

c. 10 : 00 am, 10 : 30 pm, 11 : 00 am, 11 : 30 pm, noon.

d. Until 1 : 45 am Sunday, 8 am to 2 : 15 pm and after 8 : 30 pm.

16. a. $x = 30°, 210°$ **b.** $x = 153.43°, 333.43°$

17. $x = 30°, 150°$

18. a. 100^g **b.** 200^g **c.** 300^g **d.** 400^g

Project

1. Follow the given instructions.

2. a. 4π cm

b. 3π cm

c. π cm

d. The largest arc length equals the sum of the two smaller arc lengths.

3. Sample responses can be found in the worked solutions in the online resources.

4. a. 3π cm^2

b. In ΔAYZ: $AZ^2 = AY^2 + YZ^2$

$= 6^2 + YZ^2$

In ΔBYZ: $BZ^2 = BY^2 + YZ^2$

$= 2^2 + YZ^2$

Adding these equations: $AZ^2 + BZ^2 = 6^2 + YZ^2 + 2^2 + YZ^2$

But in ΔAZB: $AZ^2 + BZ^2 = AB^2$

$6^2 + YZ^2 + 2^2 + YZ^2 = 8^2$

$2YZ^2 = 64 - 36 - 4$

$YZ^2 = 12$

$YZ = \pm\sqrt{12}$

But $YZ > 0$ as it is a length: $YZ = 2\sqrt{3}$

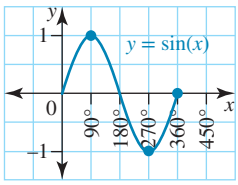
c. 3π cm^2

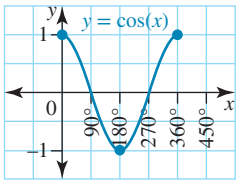
Yes, same area

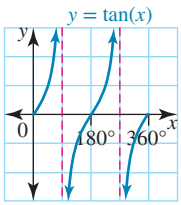
5. Area of the arbelos $= \pi r_1 r_2$

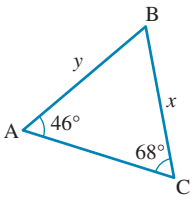
Sample responses can be found in the worked solutions in the online resources.

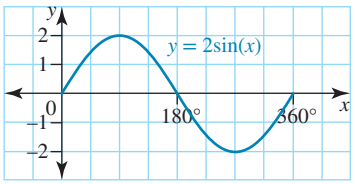
15.8 Review questions

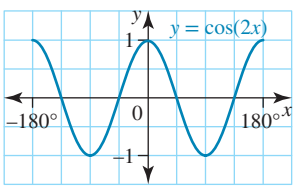
1. 14.15 cm
2. $20°31'$
3. $b = 22.11\text{ m}, c = 5.01\text{ m}, \text{C} = 10°$
4. 3.64 cm
5. 34°
6. $94°56'$
7. **a.** 159.10 cm^2 **b.** 17.68 cm
8. 4th quadrant
9. **a.** 0.94, −0.34 **b.** −2.75
10. B
11. tan(53°)
12. 

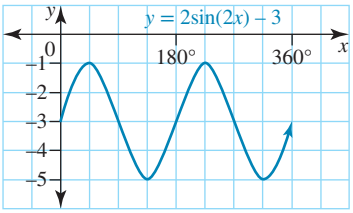
13. 

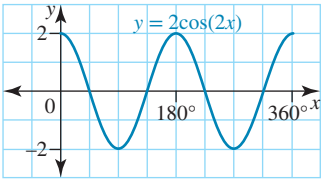
14. 

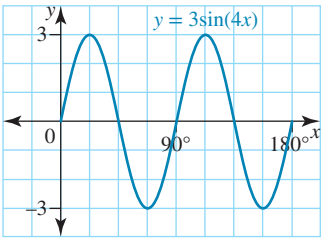
15. 

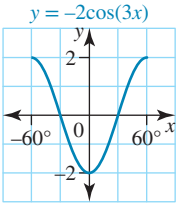
16. **a.** Period = 120°, amplitude = 2
    b. Period = 180°, amplitude = 3
    c. Period = 180°, amplitude = 1
17. **a.** 

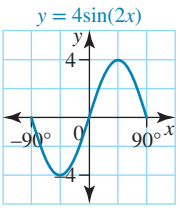
    b. 

18. **a.** $x = 191.54, 348.46$
    b. $x = 22.79, 157.21, 202.79, 337.21$
    c. $x = 88.09, 271.91$
    d. $x = 7.02, 97.02, 187.02, 277.02$
19. **a.** 30°, 150° **b.** 30°, 330°
    c. 45°, 315° **d.** 45°, 135°
20. D
21. **a.** 

    b. Period = 180, amplitude = 2
22. **a.** 

    i. Period = 180°
    ii. Amplitude = 2
    b. 

    i. Period = 90°
    ii. Amplitude = 3
    c. 

    i. Period = 120°
    ii. Amplitude = 2
    d. 

    i. Period = 180°
    ii. Amplitude = 4

23. **a.** 15°, 165°, 195°, 345°

b. −70°, 10°, 50°

c. 22.5°, 67.5°, 202.5°, 247.5°

d. 45°, 75°, 165°, 195°, 285°, 315°

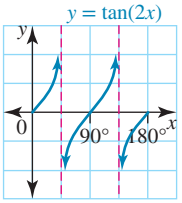
e. 0°, 45°, 90°, 135°, 180°

f. 33.75°, 78.75°, 123.75°, 168.75°

24. **a.** 60°, 300° **b.** 240°, 300°

c. 45°, 315° **d.** 225°, 315°

25.

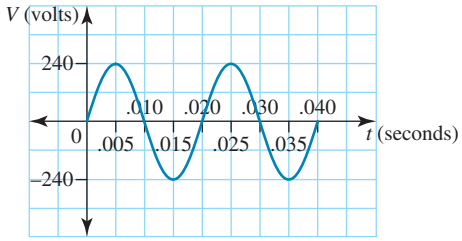


Period = 90°; amplitude is undefined.
Asymptotes are at $x = 45°$ and $x = 135°$.

26. 3.92 m

27. **a.**

t	*V*
0.000	0
0.005	240
0.010	0
0.015	−240
0.020	0
0.025	240
0.030	0
0.035	−240
0.040	0



b. Maximum voltage occurs at $t = 0.005$ s, 0.025 s.

c. 0.02 s

d. 50 cycles per second